

The Cost of Following the Central Path

Abstract

Interior-point methods take bounded steps in the local Hessian norm of a self-concordant barrier; up to constants, the least number of such steps between two points is their distance in the Hessian metric. We ask how much longer the central path is, in this metric, than the shortest route from the analytic center to a point whose objective value is within ε of the optimum. For boxes, spectral-norm balls of real and complex matrices, spectral intervals of Euclidean Jordan algebras, and products of these domains, with their standard logarithmic and log-determinant barriers, we compute these distances exactly. If all factors have the same barrier coefficient and the linear objective has r nonzero singular values or eigenvalues (coefficients, for a box), every arc of the central path has length at most $\Gamma_r = [\sum_{i=1}^r (\sqrt{i} - \sqrt{i-1})^2]^{1/2}$ times the distance between its endpoints, where $\Gamma_r^2 = \frac{1}{4} \log r + O(1)$, and no smaller constant works. The central path to its first ε -accurate point has length at most $c_* \Gamma_r$ times the distance to the set of ε -accurate points, where $1.37486 < c_* < 1.38$. For an explicit sparse linear program on $(-1, 1)^r$ with rational data and tolerance $\varepsilon_r = e^{-\Theta(r)}$, the least number of bounded steps from the analytic center to an ε_r -accurate point is $\Theta(r\sqrt{\log r})$, but $\Theta(r \log r)$ if every iterate must stay within a fixed Hessian-metric distance of the central path. The same gap, a factor $\Theta(\sqrt{\log r})$, holds for the universal and entropic barriers of the cube and, uniformly in its parameters, for a family of non-separable barriers on the cube with the least possible barrier parameter r . On the same program, reaching a strictly feasible primal–dual pair with small duality gap takes $\Theta(r^{3/2})$ steps in the primal–dual barrier metric. We count bounded steps, not arithmetic or oracle cost.

Keywords: interior-point methods; self-concordant barriers; central path; Riemannian distance; spectral optimization.

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1 Introduction

Path-following interior-point methods approach an optimum along the central path of a self-concordant barrier F . Each step is short in the local norm defined by the Hessian of F : a step from x to y is allowed if $\|y - x\|_{F,x} \leq R$, for a fixed $0 < R < 1$. Up to integer rounding and constants that depend only on R , the least number of such steps between two points equals their distance in the Riemannian metric defined by this Hessian ([corollary 2.2](#)). We ask how much longer the central path is, in this metric, than the shortest route that achieves the same goal.

The answer depends on the goal. A competing route may have to reach the same central point, reach any point with the same objective accuracy, or end at a feasible primal–dual pair with a small duality gap. Which routes are allowed also matters. For the sparse linear program on the cube $(-1, 1)^r$ of [section 6](#), the least number of steps takes three different orders:

$$\underbrace{r\sqrt{\log r}}_{\text{any primal route}}, \quad \underbrace{r \log r}_{\text{near the central path}}, \quad \underbrace{r^{3/2}}_{\text{any feasible primal–dual route}}.$$

Items 2 and 4 below give the precise conditions.

These are geometric comparisons, not operation counts: a linear objective on a box is optimized by reading the signs of its coefficients. The examples measure the cost of requiring short steps in the barrier metric. They give no lower bound for algorithms without this requirement and no arithmetic or oracle advantage for any method.

1.1 Why the central path can be long

The standard logarithmic barrier of the cube is $U(x) = \sum_i b(x_i)$ on $(-1, 1)^r$, where $b(t) = -\log(1 - t^2)$. In the coordinates $y_i = \rho(x_i)$, where $\rho(t) = \int_0^t \sqrt{b''(u)} du$, its Hessian metric is Euclidean and shortest paths are straight lines. For a linear objective with positive weights, the central path in these coordinates is $y_i(s) = H(s - a_i)$ for one fixed increasing function H . Here s is the logarithm of the central-path parameter, the shift a_i is determined by the i th objective weight, and the common speed H' increases from zero to one. When the shifts are spread out, groups of coordinates start to move one after another, so the central path turns through successive coordinate directions, whereas the shortest route moves all coordinates together ([figure 1](#)).

1.2 Main results

We write d_F for distance in the Hessian metric of F . The objective gap of a point is the optimal value minus its objective value. We write $L_{\text{opt}}(\varepsilon)$ for the distance from the analytic center to the set of points with gap at most ε (the ε -accurate set), and $L_{\text{CP}}(\varepsilon)$ for the length of the central path from the analytic center to its first point with gap ε .

1. *Exact distances and the sharp rank factor.* Consider spectral-norm balls of real or complex rectangular matrices, spectral intervals in any Euclidean Jordan algebra, and products of these domains, with their standard log-determinant barriers. The distance from the analytic center is the Euclidean norm of the singular values or eigenvalues after applying ρ to each of them; paths that rotate the spectral frame are not shorter. The squared distance $L_{\text{opt}}(\varepsilon)^2$ is the value of a strictly convex problem in the spectral coordinates, whose solution is determined by a single Lagrange multiplier ([theorems 3.1 and 3.3](#)). Suppose that all factors have the same scale (the coefficient of their barrier) and that the objective has r nonzero singular values or eigenvalues.

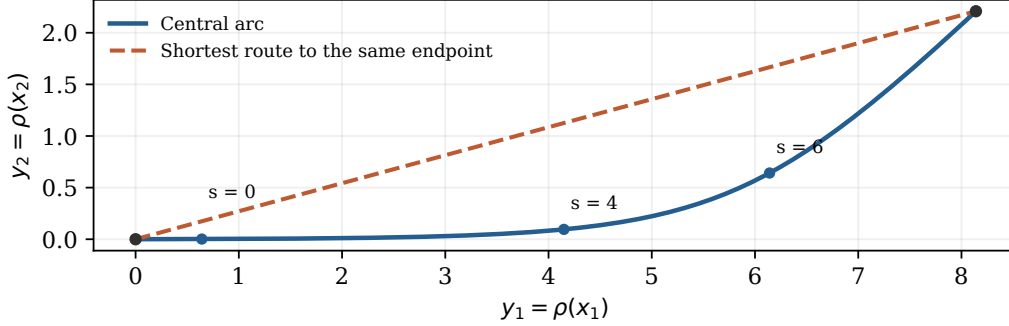


Figure 1: A central path in the coordinates $y_i = \rho(x_i)$ for the barrier $b(x_1) + b(x_2)$ and objective weights $(1, e^{-6})$, from the analytic center ($s = -\infty$) to the logarithmic parameter $s = 8$. The straight segment is the shortest route to the same endpoint; the closest accurate point is not shown.

Then the length L_{CP} of every arc of the central path satisfies

$$L_{CP} \leq \Gamma_r d_F(\text{its endpoints}), \quad \Gamma_r = \left[\sum_{i=1}^r (\sqrt{i} - \sqrt{i-1})^2 \right]^{1/2}, \quad \Gamma_r^2 = \frac{1}{4} \log r + O(1), \quad (1.1)$$

and no smaller constant holds for all such objectives and arcs. For the same accuracy ε , $L_{CP}(\varepsilon) \leq c_\star \Gamma_r L_{\text{opt}}(\varepsilon)$, where $c_\star$ is the best constant in a one-dimensional inequality; we prove $1.37486 < c_\star < 1.38$ with exact rational arithmetic (theorems 4.1 and 4.3). We do not claim that the product $c_\star \Gamma_r$ is sharp. With unequal scales, the best constant depends on the order in which coordinates start to move. It can be of order $\sqrt{r}$, but only when the ratio of the largest to the smallest scale is exponential in r (proposition 4.2).

2. *Step sequences near the central path.* For an explicit objective on the cube whose weights are powers of two (the dyadic objective), an explicit tolerance ε_r , and rational input of length $\Theta(r^2)$, we have $L_{\text{opt}} = \Theta(r\sqrt{\log r})$ and $L_{CP} = \Theta(r \log r)$ (theorem 6.1). Since a step may cut across a long part of a curve, arclength alone does not bound the number of steps, so we prove directly that every sequence of steps of local size at most R that stays within a fixed distance of the central path needs $\Omega(r \log r)$ steps to reach accuracy ε_r , even if the nearby central points occur in any order; following the path achieves $O(r \log r)$. If the distance δ_r from the path grows with r , the ratio of this step count to the optimal one still tends to infinity whenever $\delta_r = o((\log r)^{1/3})$ (theorem 6.2). This covers the usual Newton-decrement neighborhoods of fixed radius $0 \leq \beta < 1/2$.
3. *Other barriers.* Let f be a fixed self-concordant barrier on an interval with $(f')^2 \leq f''$ on the side of its analytic center toward the optimum. For objectives with positive weights, the separable barrier $\sum_i f(x_i)$ has the same sharp factor Γ_r , satisfies $L_{CP}(\varepsilon) \leq C_{\text{sc}} \Gamma_r L_{\text{opt}}(\varepsilon)$ with a constant $C_{\text{sc}} < 2$ that does not depend on r , and shows the same step-count gap on the dyadic example. This class contains the universal and entropic barriers of the cube (theorem 7.1 and proposition 7.4). Without the condition $(f')^2 \leq f''$, the ratio can approach $\sqrt{r}$ for scalar barriers whose parameter grows. If a convex coupling term added to U has first and second derivatives bounded on the whole facet collar $\{x_1 \geq 1 - \delta\}$, $0 < \delta < 1$, then the worst-case ratios

L_{CP}/d_F and $L_{\text{CP}}(\varepsilon)/L_{\text{opt}}(\varepsilon)$ are still at least Γ_r ([theorem 8.1](#)). We give a coupled barrier with a dense Hessian and a coupled barrier invariant under all signed coordinate permutations, both with barrier parameter r , the least possible for the cube. For the second, the explicit radial family

$$F_{\lambda,c}(x) = U(x) - \lambda \log(c - \|x\|^2), \quad \lambda \geq 1, \quad c - r \geq 4\lambda,$$

we also prove a matching upper bound $O(\Gamma_r)$, and the separation for step sequences holds uniformly, even when λ and c depend on r ([theorem 8.3](#) and [corollary 8.4](#)). Hence neither an optimal barrier parameter nor a large symmetry group guarantees a bounded ratio. For general couplings we prove only the lower bound.

4. *The cost of a dual solution.* It is classical that, along the primal–dual central path, the speed in the combined metric of a logarithmically homogeneous barrier and its conjugate is $\sqrt{\nu}$ in the logarithmic parameter. We compute how the objective splits this speed between the primal and dual parts. For the classical barrier of the cone over a product of k balls, whose parameter $\nu = k + 1$ is optimal among logarithmically homogeneous barriers on this cone, the primal part can be much smaller than $\sqrt{\nu}$, even when all objective weights are positive and the optimizer is unique. Write the dyadic linear program in standard form with the orthant barrier. From its central point with $s = 0$ to the central point with $s = T = \Theta(r)$, which is ε_r -accurate, the least number of primal steps is again $\Theta(r\sqrt{\log r})$, or $\Theta(r \log r)$ through exact central points. In the primal–dual barrier metric, the least number of steps to a strictly feasible primal–dual pair with gap at most $2\varepsilon_r$ is $\Theta(r^{3/2})$; the lower bound holds even if intermediate points violate the equality constraints ([theorem 9.3](#)).

For any distribution of the objective’s singular values or eigenvalues, we also give explicit two-sided bounds on the central speed and on the objective gap along the path, with geometric and polynomial decay as examples; a simpler lower bound on $L_{\text{opt}}(\varepsilon)$, based on determinants, can be zero while $L_{\text{opt}}(\varepsilon)$ tends to infinity ([section 5](#)). For reformulations with auxiliary variables over products of symmetric cones, with log-determinant barriers of coefficients $\alpha_i \geq 1$, the rank of a dual solution that exposes the optimum bounds primal distance from below, even when the auxiliary variables move freely. A bound on the dimensions of the cone factors bounds primal–dual movement from below for every logarithmically homogeneous barrier. Grouped balls, norm trees, and packings of positive semidefinite columns are worked out completely, including exact parameters of the restricted barriers and balls of different sizes with unequal objective weights ([section 10](#) and [appendix C](#)).

1.3 Relation to prior work

Nesterov and Todd [[21](#)] developed the Riemannian geometry of self-concordant barriers, its relation to short steps, and the rule that squared distances add over products. Their [Section 6.2](#) flattens the metric of a separable barrier on the hypercube by integrating its scalar local norm. Their primal–dual theorem bounds the central-path length by $\sqrt{2}$ times the endpoint distance in the metric of a logarithmically homogeneous cone barrier plus its conjugate, and their [Theorem 5.1\(c\)](#) and [Corollary 5.1](#) extend this bound to the whole set of feasible pairs with small gap, allowing intermediate points to violate the affine constraints. We restate this known result as [theorem 9.2](#) for contrast with shorter primal routes.

Nesterov and Nemirovski [[19](#)] compared primal central paths with geodesics. If D is the distance between two central points, their [Theorem 4.1](#) gives

$$L \leq \log 2 + \nu^{1/4} \sqrt{D(D + \log 12)}. \tag{1.2}$$

Their Example 1.1 distinguishes a central endpoint from a set of accurate points, and their Theorem 5.3 compares with such a set under an explicit separation assumption. On specific spectral domains, we compute the distance to the set of accurate points exactly and replace the dependence on the barrier parameter ν by an exact constant that depends only on the objective rank.

Worst-case central paths and lower bounds for path-following methods have a separate history. Deza, Nematollahi and Terlaky [9] prove iteration lower bounds with Klee–Minty constructions. Allamigeon, Benchimol, Gaubert and Joswig [1] construct long, winding logarithmic central paths with exponential iteration lower bounds for methods that stay near the path, and Allamigeon, Gaubert and Vandame [2] prove exponential lower bounds for self-concordant path following in their neighborhood model. Allamigeon, Dadush, Loho, Natura and Véggh [3] compare interior-point methods, across self-concordant barriers, with *straight-line complexity*, the number of affine segments needed in a wide neighborhood of the central path; Dadush, Ma, Natura and Véggh [8] improve these wide-neighborhood bounds and prove that trust-region path following in the narrow ℓ_2 neighborhood is approximately instance optimal. These works count affine segments or iterations, not the fixed-radius steps counted here. Our dyadic example separates movement near the central path from unrestricted movement in the same metric on a simple domain; it is neither the first example of an inefficient central path nor an exponential lower bound for every barrier.

Several ingredients are classical. The prefix inequality behind Γ_r is a finite weighted Lorentz-norm inequality [18]; what is new is that central-path velocities come arbitrarily close to it. The Hessian calculations for matrices and Jordan algebras use spectral differentiation [4, 17]. Ohara, Ishi and Tsuchiya [22] give metric projections and Hessian isometries between conjugate barriers. The universal and entropic barriers and their exact parameters are known [5, 7, 15, 20]; we derive their central-path consequences. Parameter-preserving diagonal quadratic regularization of box barriers is also known [6]; for our coupled examples, which also have the exact parameter, we add central-path comparisons.

Geodesic steps in symmetric-cone algorithms [24] and interior-point methods on Riemannian manifolds [14] address different settings. The real Hessian metric on a bounded spectral ball used here differs from the affine-invariant metric on a symmetric cone and from complex Bergman metrics.

To our knowledge, the following are new: the exact supremum (1.1) and, combined with the exact distance to the accurate set, the bound $L_{\text{CP}}(\varepsilon) \leq c_\star \Gamma_r L_{\text{opt}}(\varepsilon)$; the gap between $\Theta(r\sqrt{\log r})$ and $\Theta(r \log r)$ steps for sequences whose nearby central points may occur in any order; the comparisons for the radial family, uniform in λ and c ; and the three step-count orders on one linear program.

1.4 Organization

Section 2 defines the metric and the step counts. Sections 3 and 4 compute exact distances and prove the sharp comparison. Sections 5 and 6 treat the distribution of objective weights and step sequences, sections 7 and 8 other separable and coupled barriers, section 9 primal versus primal–dual step counts, and section 10 other formulations. The appendices contain the rational certificate for $c_\star$, the proof that $C_{\text{sc}} < 2$, a smooth barrier with nonmonotone central speed, and the proofs for the concrete formulations. The figures illustrate proved results numerically and can be reproduced with the included scripts.

2 Barrier geometry and the comparison problem

The paper makes three different comparisons: a central arc with the shortest path to the same endpoint; a central arc with the shortest path to any point that is at least as accurate; and primal movement with the movement of a feasible primal–dual pair. Each statement specifies its set of endpoints and its metric.

2.1 Definitions and step counts

Let Ω be an open convex subset of a finite-dimensional real inner product space E . A C^3 function $F : \Omega \rightarrow \mathbb{R}$ with positive definite Hessian defines a local norm, curve length, and distance:

$$\|h\|_{F,x}^2 = D^2F(x)[h, h], \quad \text{len}_F(\gamma) = \int_0^1 \|\dot{\gamma}(t)\|_{F,\gamma(t)} dt, \quad d_F(x, y) = \inf_{\gamma: x \rightsquigarrow y} \text{len}_F(\gamma). \quad (2.1)$$

The infimum is over absolutely continuous curves in Ω . On an affine slice of Ω , norms are restricted to the tangent space of the slice, and curves stay in the slice. Complex matrix spaces are treated as real vector spaces, with inner product $\text{Re tr}(X^*Y)$.

We call F a standard self-concordant barrier if it tends to $+\infty$ at every finite boundary point and satisfies

$$|D^3F(x)[h, h, h]| \leq 2\|h\|_{F,x}^3. \quad (2.2)$$

Its barrier parameter is any ν satisfying $|DF(x)[h]| \leq \sqrt{\nu}\|h\|_{F,x}$ for all x, h ; the least such ν is the exact parameter. In “standard self-concordant”, the word “standard” refers to the constant 2 in (2.2); “the standard barriers” means the logarithmic and log-determinant barriers of [section 3](#) and their products, with any positive scales. They are standard self-concordant when every scale is at least one ([proposition 3.2](#)). Self-concordance implies the polarized inequality

$$|D^3F(x)[u, h, h]| \leq 2\|u\|_{F,x}\|h\|_{F,x}^2. \quad (2.3)$$

We write $\|h\|_x$ when F is fixed.

Lemma 2.1 (Distance and the local norm of a chord). *For a standard self-concordant barrier,*

$$\log(1 + \|y - x\|_x) \leq d_F(x, y). \quad (2.4)$$

If $q = \|y - x\|_x < 1$, the straight segment from x to y has length at most $-\log(1 - q)$. In particular, for $0 < R < 1$,

$$\|y - x\|_x \leq R \implies d_F(x, y) \leq -\log(1 - R), \quad d_F(x, y) \leq \log(1 + R) \implies \|y - x\|_x \leq R. \quad (2.5)$$

Proof. Along any curve γ starting at x , let $\ell(t) = \int_0^t \|\dot{\gamma}(u)\|_{\gamma(u)} du$. For a fixed vector $h \neq 0$, (2.3) gives $|\frac{d}{dt} \log\|h\|_{\gamma(t)}| \leq \dot{\ell}(t)$ almost everywhere. Thus $\|h\|_x \leq e^{\ell(t)}\|h\|_{\gamma(t)}$ for every h . The triangle inequality in the fixed norm at x now gives

$$\|y - x\|_x \leq \int_0^1 \|\dot{\gamma}(t)\|_x dt \leq \int_0^1 e^{\ell(t)} \dot{\ell}(t) dt = e^{\text{len}_F(\gamma)} - 1.$$

Taking the infimum proves (2.4). For the segment $x + th$, $h = y - x$, set $a(t) = \|h\|_{x+th}$. Self-concordance gives $|a'(t)| \leq a(t)^2$ and hence $a(t) \leq q/(1 - tq)$ while $tq < 1$. Integration from 0 to 1 gives the second assertion. The segment lies in Ω by convexity. $\square$

These estimates underlie the short-step counts of Nesterov and Todd [21, Section 3]. For a target set $A \subset \Omega$ and a starting point x_0 , let $D_F(x_0, A) = \inf_{x \in A} d_F(x_0, x)$. A forward- R sequence is a finite sequence $x_0, x_1, \dots, x_T$ of points in Ω such that $\|x_{j+1} - x_j\|_{x_j} \leq R$; its T steps are called forward R -chords.

Corollary 2.2 (Optimal bounded movement). *Let F be a standard self-concordant barrier and $0 < R < 1$. Every forward- R sequence from x_0 to A satisfies*

$$T \geq \frac{D_F(x_0, A)}{-\log(1 - R)}. \quad (2.6)$$

If a shortest curve to A exists, there is such a sequence with

$$T \leq \left\lceil \frac{D_F(x_0, A)}{\log(1 + R)} \right\rceil. \quad (2.7)$$

Proof. For the lower bound, join consecutive points by straight segments and apply lemma 2.1. For the upper bound, divide a shortest curve into arcs of length at most $\log(1 + R)$ and apply (2.4) to their endpoints. If the distance is zero and attained, no step is needed. $\square$

For a bounded domain and a nonzero objective vector $c \in E$, define the support value, the objective gap, and the ε -accurate set:

$$h_\Omega(c) = \sup_{x \in \Omega} \langle c, x \rangle, \quad g_c(x) = h_\Omega(c) - \langle c, x \rangle, \quad A_\varepsilon = \{x \in \Omega : g_c(x) \leq \varepsilon\}. \quad (2.8)$$

When F is a strictly convex barrier, the central point with parameter η is

$$x(\eta) = \arg \min_{x \in \Omega} \{F(x) - \eta \langle c, x \rangle\}, \quad \eta \geq 0. \quad (2.9)$$

At $\eta = 0$ it is the analytic center x_0 . We write $L_{\text{opt}}(\varepsilon) = D_F(x_0, A_\varepsilon)$ and $L_{\text{CP}}(\eta) = \text{len}_F(x|_{[0, \eta]})$; $L_{\text{CP}}(\eta_0, \eta_1)$ is the length of the arc between parameters η_0 and η_1 , and $L_{\text{CP}}(\varepsilon) = L_{\text{CP}}(\eta_\varepsilon)$, where η_ε is the first parameter with $g_c(x(\eta_\varepsilon)) = \varepsilon$. We say whether the argument of L_{CP} is a parameter or an accuracy whenever this is ambiguous.

3 Exact distance and the closest accurate endpoint

All distances in this section are expressed through the scalar coordinate ρ , which makes the metric of b Euclidean. Let

$$b(t) = -\log(1 - t^2), \quad \rho(t) = \int_0^t \frac{\sqrt{2(1 + u^2)}}{1 - u^2} du, \quad -1 < t < 1. \quad (3.1)$$

Then ρ is odd, strictly increasing, maps $(-1, 1)$ onto $\mathbb{R}$, and $\rho'(t)^2 = b''(t)$. For $0 \leq t < 1$,

$$\log \frac{1}{1 - t} \leq \rho(t) \leq \sqrt{2} \log \frac{1}{1 - t}. \quad (3.2)$$

Indeed, after multiplying the corresponding derivative inequalities by $1 - t$, these bounds reduce to $1 \leq \sqrt{2(1 + t^2)}/(1 + t) \leq \sqrt{2}$.

3.1 Boxes, rectangular matrix balls, and Jordan intervals

For a box $(-1, 1)^n$ with $F(x) = \sum_i \alpha_i b(x_i)$ and $\alpha_i > 0$, integrating the diagonal metric gives

$$d_F(x, y)^2 = \sum_i \alpha_i (\rho(x_i) - \rho(y_i))^2. \quad (3.3)$$

The straight segment in the coordinates $\rho(x_i)$ stays in the box and attains this distance. This construction is classical. The next theorem shows that, on matrix balls and Jordan intervals, leaving a spectral frame does not shorten a path from the center.

For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $p \leq q$, put

$$\mathcal{B}_{p,q}^{\mathbb{F}} = \{X \in \mathbb{F}^{p \times q} : \|X\|_{\text{op}} < 1\}, \quad \Phi_\alpha(X) = -\alpha \log \det(I_p - XX^*), \quad \alpha > 0. \quad (3.4)$$

Write $\sigma^\downarrow(X)$ for the p singular values of X in decreasing order, including zeros.

We also use a Euclidean Jordan algebra V , of rank n , with unit e , cone of squares K , Jordan product $x \circ y$, and trace inner product $\langle x, y \rangle = \text{tr}(x \circ y)$. A primitive idempotent is a nonzero element u with $u \circ u = u$ that cannot be written as the sum of two nonzero orthogonal idempotents. A Jordan frame $(e_i)_{i=1}^n$ consists of orthogonal primitive idempotents with sum e . The spectral theorem gives $x = \sum_i \lambda_i(x) e_i$ and $\det x = \prod_i \lambda_i(x)$. The Jordan interval and its barrier are

$$\mathcal{I}(V) = \{x : -e \prec_K x \prec_K e\}, \quad \Phi_\alpha(x) = -\alpha \log \det(e - x^2) = \alpha \sum_{i=1}^n b(\lambda_i(x)). \quad (3.5)$$

Here $x^2 = x \circ x$ and $u \prec_K v$ means $v - u \in \text{int } K$. Examples are real, complex, and quaternionic Hermitian matrices, Lorentz algebras, their direct products, and the exceptional Albert algebra. The number of spectral coordinates is the rank n , not the real dimension of V .

Theorem 3.1 (Exact spectral distance). *For every $\alpha > 0$, the following identities hold:*

$$d_{\Phi_\alpha}(0, X)^2 = \alpha \sum_{i=1}^p \rho(\sigma_i(X))^2, \quad X \in \mathcal{B}_{p,q}^{\mathbb{F}}, \quad (3.6)$$

$$d_{\Phi_\alpha}(0, x)^2 = \alpha \sum_{i=1}^n \rho(\lambda_i(x))^2, \quad x \in \mathcal{I}(V). \quad (3.7)$$

For general endpoints,

$$d_{\Phi_\alpha}(X, Y) \geq \sqrt{\alpha} \|\rho(\sigma^\downarrow(X)) - \rho(\sigma^\downarrow(Y))\|_2, \quad (3.8)$$

$$d_{\Phi_\alpha}(x, y) \geq \sqrt{\alpha} \|\rho(\lambda^\downarrow(x)) - \rho(\lambda^\downarrow(y))\|_2. \quad (3.9)$$

Here ρ acts coordinatewise, and the Jordan eigenvalues are ordered by signed value. Equality holds in both lower bounds when the endpoints have a common spectral frame in which both spectra are ordered. The minimizing path is then linear in the ρ coordinates of that frame.

Proof. First consider matrices. Introduce the Hermitian dilation and a scalar function

$$S(X) = \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}, \quad g(t) = -\frac{1}{2} \log(1 - t^2).$$

The eigenvalues of $S(X)$ are $\pm \sigma_i(X)$ and $q - p$ zeros, so $\Phi_\alpha(X) = \alpha \text{tr } g(S(X))$. In an eigenbasis of a Hermitian matrix S with eigenvalues λ_a , the Hessian of $\text{tr } g(S)$ is

$$D^2 \text{tr } g(S)[H, H] = \sum_a g''(\lambda_a) |H_{aa}|^2 + 2 \sum_{a < b} \frac{g'(\lambda_a) - g'(\lambda_b)}{\lambda_a - \lambda_b} |H_{ab}|^2. \quad (3.10)$$

The quotient at equal eigenvalues means $g''(\lambda_a)$. For completeness, the gradient of $\text{tr } g(S)$ is $g'(S)$. Differentiating a polynomial in S shows that the (a, b) entry of $Dg'(S)[H]$ is the displayed divided difference times H_{ab} . The convergent power series of g on $(-1, 1)$ justifies the same calculation here, uniformly on compact spectral subintervals. Taking the inner product of $Dg'(S)[H]$ with H gives (3.10), over the real and complex fields alike. This is the trace-separable case of the spectral Hessian formula [17].

All divided differences in (3.10) are positive. The ordered eigenvalues of an absolutely continuous Hermitian path are absolutely continuous; almost everywhere their derivatives are the eigenvalues of the derivative compressed to each repeated eigenspace. Thus one can choose the eigenbasis within each such space to identify the diagonal contributions with these derivatives. Applying the formula to $S(X(t))$ gives

$$\|\dot{X}(t)\|_{\Phi_\alpha, X(t)}^2 \geq \alpha \sum_i b''(\sigma_i(X(t))) \dot{\sigma}_i(X(t))^2 = \alpha \left\| \frac{d}{dt} \rho(\sigma(X(t))) \right\|_2^2 \quad \text{a.e.} \quad (3.11)$$

The factor is b'' because each pair $\pm \sigma_i$ contributes $2g'' = b''$. Singular values are Lipschitz functions of the matrix, and at zero or repeated singular values the same almost-everywhere rule follows from the dilation. Integration proves (3.8). For a fixed singular value decomposition $X = U \text{diag}(\sigma_i) V^*$, the path

$$X(t) = U \text{diag}(\rho^{-1}(t\rho(\sigma_i))) V^*, \quad 0 \leq t \leq 1, \quad (3.12)$$

has constant speed equal to the square root of the right side of (3.6). Indeed, restricting Φ_α to this frame gives $\alpha \sum_i b(x_i)$. Therefore the lower bound is attained. Linear interpolation between two transformed spectra ordered in a common frame preserves the order and gives the equality case of (3.8).

Now consider Jordan algebras. Relative to a Jordan frame, the Peirce spaces are $V_{ii} = \mathbb{R}e_i$ and, for $i < j$,

$$V_{ij} = \{h : e_i \circ h = e_j \circ h = \tfrac{1}{2}h, \quad e_k \circ h = 0 \text{ for } k \notin \{i, j\}\}.$$

At $x = \sum_i \lambda_i e_i$, write the orthogonal Peirce decomposition of a vector h as $h = \sum_i h_i e_i + \sum_{i < j} h_{ij}$, where $h_{ij} \in V_{ij}$ and $\|e_i\| = 1$. The Hessian is

$$D^2\Phi_\alpha(x)[h, h] = \alpha \sum_i b''(\lambda_i) h_i^2 + \alpha \sum_{i < j} \frac{b'(\lambda_i) - b'(\lambda_j)}{\lambda_i - \lambda_j} \|h_{ij}\|^2. \quad (3.13)$$

To verify this formula for every type of Jordan algebra, differentiate the gradient $\alpha((e - x)^{-1} - (e + x)^{-1})$. To compute the derivative of inversion at a positive diagonal element $z = \sum_i z_i e_i$, let $L_z h = z \circ h$ and differentiate $z \circ z^{-1} = e$. This gives $L_z D(z^{-1})[h] = -L_{z^{-1}} h$. The operator L_z is positive definite: it acts by z_i on V_{ii} and by $(z_i + z_j)/2$ on V_{ij} . On the same spaces, $L_{z^{-1}}$ acts by z_i^{-1} and $(z_i^{-1} + z_j^{-1})/2$, respectively. Solving the differentiated identity therefore gives the inversion coefficients $-z_i^{-2}$ and $-(z_i z_j)^{-1}$. The resulting off-diagonal coefficient is

$$\frac{1}{(1 - \lambda_i)(1 - \lambda_j)} + \frac{1}{(1 + \lambda_i)(1 + \lambda_j)} = \frac{b'(\lambda_i) - b'(\lambda_j)}{\lambda_i - \lambda_j}.$$

This also gives its continuous value at a repeated eigenvalue. Every coefficient is positive. Jordan eigenvalues obey the same local Lipschitz and repeated-eigenvalue derivative rule as matrices: within the Peirce algebra for a repeated eigenvalue, diagonalize the corresponding component of h . Equivalently, this follows from the spectral variational principle applied to $x + th$. Hence, along an absolutely continuous curve,

$$\|\dot{x}\|_{\Phi_\alpha, x}^2 \geq \alpha \sum_i b''(\lambda_i(x)) \dot{\lambda}_i(x)^2.$$

Integrate and interpolate in a fixed Jordan frame exactly as above. The argument uses only the Jordan spectral theorem and the Peirce rules, so it covers the exceptional algebra. $\square$

The theorem needs only $\alpha > 0$, because distance depends only on the Hessian. Conclusions about local steps also need self-concordance.

Proposition 3.2 (Self-concordance and parameter). *If $\alpha \geq 1$, the barriers in (3.4) and (3.5) are standard self-concordant barriers. Their least barrier parameters are αp and αn , respectively. For a product with scales $\alpha_j \geq 1$, the least barrier parameter is the sum of those of the factors.*

Proof. The matrix barrier at scale one is the restriction of the positive Hermitian log-determinant barrier to $\begin{pmatrix} I_p & X \\ X^* & I_q \end{pmatrix}$. The Jordan barrier is the restriction of $-\log \det u - \log \det v$ to $(u, v) = (e - x, e + x)$. The log-determinant is self-concordant: after a linear change of variables making its Hessian at a point the identity, the second and third directional derivatives are $\sum_i \xi_i^2$ and $-2 \sum_i \xi_i^3$, so $|\sum_i \xi_i^3| \leq (\sum_i \xi_i^2)^{3/2}$. Affine restriction and addition preserve self-concordance, as does multiplication by $\alpha \geq 1$. The barriers tend to infinity at the boundary because the determinant vanishes there.

The gradient has no component outside the spectral frame, and the Hessian has no mixed term between that frame and its orthogonal complement. Thus the squared dual norm of the gradient is

$$\|\nabla \Phi_\alpha\|_{\Phi_{\alpha,*}}^2 = \alpha \sum_i \frac{b'(t_i)^2}{b''(t_i)} = \alpha \sum_i \frac{2t_i^2}{1+t_i^2} < \alpha n_0, \quad (3.14)$$

where $n_0 = p$ and $t_i = \sigma_i$ for a matrix ball, and $n_0 = n$ and $t_i = \lambda_i$ for a Jordan interval. In the matrix case this block structure follows either from (3.10) or by differentiating the invariant trace function in singular coordinates. Letting all t_i tend to 1 proves that the supremum is αn_0 . For products, squared gradient dual norms add, and the limits can be approached simultaneously. $\square$

3.2 Distance to the set of accurate points

Consider a finite product of the domains defined above, where factor j has scale $\alpha_j > 0$. Let c be a nonzero objective, maximized as in (2.8). Its active weights $w_i > 0$ are the positive singular values of the matrix components of c and the absolute values of the nonzero Jordan eigenvalues of its Jordan components. Pool these weights over all factors, index them by $1 \leq i \leq r$, and give weight i the scale α_i of its factor. For a box, $w_i = |c_i|$ at the nonzero coordinates of c . Let $W_0 = \sum_i w_i$. The support value $h_\Omega(c)$ equals W_0 , and the analytic center is zero.

Theorem 3.3 (The closest accurate endpoint). *For $0 < \varepsilon < W_0$, the distance from the analytic center to A_ε is*

$$L_{\text{opt}}(\varepsilon)^2 = \min \left\{ \sum_{i=1}^r \alpha_i \rho(x_i)^2 : 0 \leq x_i < 1, \quad \sum_{i=1}^r w_i(1 - x_i) \leq \varepsilon \right\}. \quad (3.15)$$

This scalar allocation problem has a unique minimizer, all its coordinates lie in $(0, 1)$, and the accuracy constraint is active. There is a unique $\lambda > 0$ such that

$$2\alpha_i \rho(x_i) \rho'(x_i) = \lambda w_i \quad (1 \leq i \leq r), \quad \sum_i w_i(1 - x_i) = \varepsilon. \quad (3.16)$$

The point in the objective's spectral frame with these active coordinates, signed like the objective, and zero inactive coordinates attains the minimum distance in the original domain. The path to it that is linear in the transformed spectral coordinates is a shortest path to A_ε .

Proof. Distances in a product metric add in squares. To see this directly, the length of a product curve is the integral of the Euclidean norm of its component speeds, which is at least the Euclidean norm of the component lengths. Constant-speed shortest component curves attain equality; such curves from zero were constructed in [theorem 3.1](#).

For a matrix factor, von Neumann's trace inequality gives $\langle C, X \rangle \leq \sum_i w_i \sigma_i(X)$, with weights and singular values ordered. Thus moving an endpoint into the objective's singular frame, keeping its singular values, cannot decrease its objective value and leaves its distance unchanged. Inactive singular values can then be set to zero, reducing the distance. For a Jordan factor, if (e_i) and (f_j) are frames of c and x , the entries $m_{ij} = \langle e_i, f_j \rangle$ are nonnegative and their row and column sums are one. Therefore

$$\langle c, x \rangle = \sum_{i,j} \mu_i \lambda_j m_{ij} \leq \sum_i |\mu_i|^\downarrow |\lambda_i|^\downarrow.$$

The last inequality is the rearrangement inequality for a doubly stochastic matrix. Assigning magnitudes to the objective frame in this order and using the signs of μ_i attains the bound at the same distance from the center. Zero-weight coordinates can again be set to zero. Conversely, every feasible vector in (3.15) gives a feasible endpoint in the objective frame with exactly that objective gap and distance. This proves the reduction.

For $x \geq 0$, $\rho' > 0$, $\rho'' \geq 0$, and $(\rho^2)'' = 2((\rho')^2 + \rho\rho'') > 0$. The objective is therefore strictly convex. It tends to infinity if any coordinate tends to 1. A nonempty bounded sublevel set is compact in $[0, 1]^r$, so a minimizer exists and is unique. Its accuracy constraint must be active: otherwise reduce one positive coordinate slightly. Slater's condition holds by taking every coordinate sufficiently close to 1. The KKT multiplier is positive, since a zero multiplier would force every coordinate to be zero, which violates $\varepsilon < W_0$. At $x_i = 0$ the objective's derivative is zero, whereas the accuracy constraint contributes $-\lambda w_i < 0$; stationarity with a nonnegative multiplier for $-x_i \leq 0$ is then impossible. Every coordinate is positive, and stationarity gives (3.16).

The function $(\rho^2)'$ increases continuously from zero to infinity. For a fixed $\lambda > 0$, each equation in (3.16) thus has one solution $x_i(\lambda)$. The resulting gap $\sum_i w_i(1 - x_i(\lambda))$ decreases continuously and strictly from W_0 to zero as λ increases from zero to infinity. This proves that λ is unique and gives a one-dimensional search for it. The minimizing path follows from [theorem 3.1](#) and the product rule. Scales within a factor are equal, so there the minimizer is ordered like the weights, and the common-frame construction applies. $\square$

Uniqueness concerns only the scalar allocation; at repeated objective singular values or eigenvalues we do not claim that the minimizing point in the original domain is unique. If $\varepsilon \geq W_0$, the analytic center is ε -accurate. For $\varepsilon = 0$, no strictly interior point is accurate, and $L_{\text{opt}}(\varepsilon) \rightarrow \infty$ as $\varepsilon \downarrow 0$.

Corollary 3.4 (Logarithmic allocation and bounded moves). *Let $0 < \varepsilon < W_0$. Define*

$$D_{\log}(\varepsilon)^2 = \min \left\{ \sum_i \alpha_i s_i^2 : s_i \geq 0, \quad \sum_i w_i e^{-s_i} \leq \varepsilon \right\}. \quad (3.17)$$

Then

$$D_{\log}(\varepsilon) \leq L_{\text{opt}}(\varepsilon) \leq \sqrt{2} D_{\log}(\varepsilon). \quad (3.18)$$

The optimizer of (3.17) is unique and satisfies

$$s_i = \mathcal{W}(\gamma w_i / \alpha_i), \quad \sum_i \alpha_i \mathcal{W}(\gamma w_i / \alpha_i) = \gamma \varepsilon, \quad \gamma > 0, \quad (3.19)$$

where $\mathcal{W}$ is the nonnegative inverse of $s \mapsto se^s$. For $0 < R < 1$ and all scales at least one, the least number $T_R^*(\varepsilon)$ of steps in a forward- R sequence from zero to A_ε satisfies

$$\frac{L_{\text{opt}}(\varepsilon)}{-\log(1-R)} \leq T_R^*(\varepsilon) \leq \left\lceil \frac{L_{\text{opt}}(\varepsilon)}{\log(1+R)} \right\rceil. \quad (3.20)$$

Proof. Set $s_i = -\log(1-x_i)$ and apply (3.2) to every feasible allocation. Strict convexity and the KKT conditions for (3.17) give $\alpha_i s_i = \gamma w_i e^{-s_i}$ with all $s_i > 0$. Inverting and summing these identities gives (3.19). Uniqueness follows, for example, by observing that $\sum_i w_i \exp[-\mathcal{W}(\gamma w_i/\alpha_i)]$ decreases strictly from W_0 to zero; the trivial root $\gamma = 0$ of the second equation in (3.19) is excluded by $\gamma > 0$. Finally, proposition 3.2 and corollary 2.2 apply, and the shortest curve was constructed in theorem 3.3. $\square$

Given the objective's spectral frame, the shortest route is explicit. Invert $(\rho^2)'$ in each coordinate and find λ by doubling and bisection; an upper end of the bracket with gap at most ε gives an ε -accurate point. Points spaced at arclength $\log(1+R)$ along the path form a forward- R sequence. Finite-precision evaluation must preserve accuracy and step size, for instance with strict margins. We make no claim about the arithmetic cost of this search or of computing the spectral frame.

4 Sharp comparison for the standard spectral barriers

We keep the setting, active weights $w_i > 0$, and scales $\alpha_i > 0$, $i = 1, \dots, r$, of section 3.2. An active channel is a positive singular value or a nonzero Jordan eigenvalue of the objective. Central points share the objective's spectral frame and signs, and vanish in zero channels. The metric statements in this section need only positive scales. Statements about standard self-concordance and bounded local steps also need $\alpha_i \geq 1$.

4.1 The speed of one central coordinate and the sharp constant

Define

$$q(z) = \frac{z}{1 + \sqrt{1+z^2}}, \quad H(s) = \rho(q(e^s)), \quad v(s) = H'(s) = \sqrt{1 - \frac{1}{\sqrt{1+e^{2s}}}}. \quad (4.1)$$

Here H is a strictly increasing bijection from $\mathbb{R}$ to $(0, \infty)$. In the objective frame, with $s = \log \eta$, the central coordinates are

$$x_i(\eta) = q(\eta w_i/\alpha_i), \quad a_i = \log(\alpha_i/w_i), \quad y_i(s) = \rho(x_i(e^s)) = H(s - a_i). \quad (4.2)$$

For a matrix factor of scale α , this is equivalently

$$X(\eta) = \frac{\eta}{\alpha} \left(I + \sqrt{I + (\eta/\alpha)^2 C C^*} \right)^{-1} C.$$

These formulas follow by minimizing each scalar function $\alpha_i b(x) - \eta w_i x$, whose minimizer solves $2\alpha_i x/(1-x^2) = \eta w_i$. The spectral trace inequalities used in theorem 3.3 show that the point in the objective frame with these coordinates minimizes over the whole domain. Strict convexity gives uniqueness, even for equal or zero objective weights. Differentiation gives $dx/ds = x(1-x^2)/(1+x^2)$ and then (4.1). In particular, v increases strictly from zero to one and

$$v(s) \leq e^s/\sqrt{2} \quad (s \leq 0), \quad 1 - v(s) \leq e^{-s} \quad (s \geq 0), \quad H(s) = s + \kappa + o(1) \quad (s \rightarrow \infty), \quad (4.3)$$

for a finite constant κ . The last assertion follows by integrating the first two bounds. They also show $H(s) \leq 1/\sqrt{2} + \max(s, 0)$.

Order the channels by nondecreasing a_i (channel i starts to move appreciably near $s = a_i$), breaking ties arbitrarily, and set

$$S_0 = 0, \quad S_i = \sum_{j \leq i} \alpha_j, \quad d_i = \sqrt{S_i} - \sqrt{S_{i-1}}, \quad \Gamma_{\alpha,a}^2 = \sum_{i=1}^r \frac{d_i^2}{\alpha_i}. \quad (4.4)$$

The value of $\Gamma_{\alpha,a}$ can depend on how tied channels are ordered; the following bound holds for every such order. The sharpness statement uses objectives with distinct a_i . For equal scales we write

$$\Gamma_r^2 = \sum_{i=1}^r (\sqrt{i} - \sqrt{i-1})^2 = \frac{1}{4} \log r + O(1). \quad (4.5)$$

Theorem 4.1 (Exact worst-case ratio for a prescribed scale order). *For every arc of the central path, including arcs that start at the analytic center,*

$$d_\Phi(x(\eta_0), x(\eta_1)) \leq L_{\text{CP}}(\eta_0, \eta_1) \leq \Gamma_{\alpha,a} d_\Phi(x(\eta_0), x(\eta_1)), \quad 0 \leq \eta_0 < \eta_1. \quad (4.6)$$

Fix an ordered vector of positive scales. The supremum of the ratio of the length of the central arc from the analytic center to the distance between its endpoints, over all final parameters and all objectives with $a_1 < \dots < a_r$, is exactly $\Gamma_{\alpha,a}$. It is already attained as a supremum on a product of scalar intervals.

Proof. For $h_1 \geq \dots \geq h_r \geq 0$, put $h_{r+1} = 0$. Writing h as a nonnegative combination of the indicator vectors of the initial index sets $\{1, \dots, j\}$ and applying the triangle inequality gives

$$\left(\sum_i \alpha_i h_i^2 \right)^{1/2} \leq \sum_j \sqrt{S_j} (h_j - h_{j+1}) = \sum_i d_i h_i. \quad (4.7)$$

Apply this to $h_i = v(s - a_i)$ and integrate over the parameter interval. Weighted Cauchy–Schwarz bounds the result by $\Gamma_{\alpha,a} (\sum_i \alpha_i (\Delta y_i)^2)^{1/2}$. The latter norm is the exact endpoint distance by [theorem 3.1](#). In each factor, all central points lie in one spectral frame of the objective, with the objective’s signs and with absolute values ordered like the objective’s, so the equality case applies to every central arc.

For sharpness set $c_i = d_i/\alpha_i = 1/(\sqrt{S_i} + \sqrt{S_{i-1}})$, a strictly decreasing sequence. For $M \rightarrow \infty$, choose $u_i = H^{-1}(Mc_i)$, $w_i = \alpha_i e^{u_i}$, and terminal $s = 0$. The endpoint distance is exactly $M\Gamma_{\alpha,a}$. Replacing each $v(s + u_i)$ by $\mathbf{1}_{s > -u_i}$ changes the integrated weighted norm on $(-\infty, 0]$ by at most $\sum_i \sqrt{\alpha_i} \int_{\mathbb{R}} |v - \mathbf{1}_{s > 0}| ds$, independent of M . For large M , all u_i are positive, and the step-function length is $\sum_i d_i u_i$. By [\(4.3\)](#), $u_i = Mc_i - \kappa + o(1)$, so this length is $M\Gamma_{\alpha,a}^2 + O_\alpha(1)$. Taking the ratio proves the supremum. Finally, $(\sqrt{i} - \sqrt{i-1})^2 = 1/(4i) + O(i^{-2})$ proves [\(4.5\)](#). $\square$

The right side of [\(4.7\)](#) is a finite weighted form of a classical Lorentz norm [\[18\]](#), and the prefix inequality is elementary; we do not claim it as new. What [theorem 4.1](#) adds is that central arcs come arbitrarily close to the resulting bound $\Gamma_{\alpha,a}$. Unlike [\(1.2\)](#), the constant for equal scales depends on the number r of active channels, not on the dimension. The objectives approaching the supremum can have weights of widely different sizes; [section 6](#) gives a construction with rational data.

Proposition 4.2 (Scale range and activation order). *For $r \geq 2$,*

$$\Gamma_{\alpha,a}^2 \leq 1 + (r-1) \tanh\left(\frac{\log(S_r/S_1)}{4(r-1)}\right) \leq \min\{r, 1 + \frac{1}{4} \log(S_r/S_1)\}. \quad (4.8)$$

Equality in the first inequality holds precisely when the cumulative ratios S_i/S_{i-1} , $i \geq 2$, are equal. Among orders of a fixed multiset of scales, the nondecreasing order maximizes Γ and the nonincreasing order minimizes it.

Proof. The first term of (4.4) is one; its i th term, $i \geq 2$, is $\tanh(\frac{1}{4} \log(S_i/S_{i-1}))$. Strict concavity of $\tanh$ on $(0, \infty)$ gives the first inequality and its equality condition. The bounds $\tanh z \leq \min(1, z)$ give the second. For the ordering assertion, interchange adjacent scales $a \leq b$ whose preceding scales sum to $S > 0$. The two logarithmic increments have common sum $T = \log((S+a+b)/S)$. Their first increments are $u = \log((S+a)/S) \leq v = \log((S+b)/S)$, and $u+v \geq T$ because $(S+a)(S+b) \geq S(S+a+b)$. Thus $|u - T/2| \leq |v - T/2|$. Concavity makes the symmetric sum $\tanh(u/4) + \tanh((T-u)/4)$ at least its value at v . If $S = 0$, the first contribution is one and the second is larger with the smaller scale first. Adjacent exchanges prove both assertions. $\square$

For example, partial sums S_i growing geometrically with a fixed ratio greater than one give $\Gamma = \Theta(\sqrt{r})$. Conversely, by (4.8), Γ of order $\sqrt{r}$ requires $\log(S_r/\alpha_{\min}) = \Omega(r)$. With $\alpha_i \geq 1$ and total scale polynomial in r , $\Gamma = O(\sqrt{\log r})$. For $r = 1$, $\Gamma = 1$ for every scale.

4.2 Comparison with the closest accurate point

Define $p(y) = b'(\rho^{-1}(y))$ for $y \geq 0$, using the functional inverse of ρ . Then $p'(y) = \rho'(\rho^{-1}(y)) > 0$ and p is strictly convex on $(0, \infty)$.

Theorem 4.3 (A scalar constant and the comparison at equal accuracy). *The least constant c such that $p(cy) \geq yp'(y)$ for all $y > 0$ is*

$$c_\star = \max_{y>0} \frac{p^{-1}(yp'(y))}{y}, \quad \frac{68743}{50000} < c_\star < \frac{69}{50}. \quad (4.9)$$

The maximum is attained at some $y > 0$. For every product considered here and every $0 < \varepsilon < \sum_i w_i$,

$$L_{\text{CP}}(\varepsilon) \leq c_\star \Gamma_{\alpha,a} L_{\text{opt}}(\varepsilon). \quad (4.10)$$

The constant $c_\star$ is sharp for the scalar inequality. We do not claim that its product with $\Gamma_{\alpha,a}$ is the sharp constant in (4.10).

Proof. Use $v = \sqrt{2}x/\sqrt{1+x^2} \in (0, 1)$ and write

$$\begin{aligned} Y(v) &= 2 \operatorname{artanh} v - \sqrt{2} \operatorname{artanh}(v/\sqrt{2}) = \rho(x), \\ A(v) &= \frac{\sqrt{2-v^2}}{1-v^2} = p'(Y(v)), \quad P(v) = vA(v) = p(Y(v)), \quad Y'(v) = \frac{2}{(1-v^2)(2-v^2)}. \end{aligned} \quad (4.11)$$

These identities follow by differentiation and equality at zero. Put

$$W(v) = \sqrt{1 - (1 + Y(v)^2 A(v)^2)^{-1/2}}.$$

The ratio defining $c_\star$ is $Y(W(v))/Y(v)$. As $v \downarrow 0$, $Y(v) = v + O(v^3)$ and this ratio tends to one. As $v \uparrow 1$, $Y(v) = \log(1/(1-v)) + O(1)$ and $1 - W(v) = \Theta((1-v)/Y(v))$, so $Y(W(v)) =$

$Y(v) + \log Y(v) + O(1)$ and the ratio again tends to one. Strict convexity gives $yp'(y) > p(y)$, so the ratio exceeds one inside the interval. Continuity and these endpoint limits prove attainment. [Appendix A](#) proves both strict rational bounds.

Let $y_i^* = \rho(x_i^*)$ be the exact allocation in [theorem 3.3](#), with multiplier $\lambda > 0$. Its equations are $2\alpha_i y_i^* p'(y_i^*) = \lambda w_i$. At central parameter $\eta = \lambda/2$, the central coordinates obey $\alpha_i p(y_i^c) = \lambda w_i/2$, hence

$$p(y_i^c) = y_i^* p'(y_i^*), \quad y_i^* \leq y_i^c \leq c_* y_i^*.$$

This central point dominates the optimal allocation, so it is ε -accurate. The first ε -accurate central point therefore has parameter at most $\lambda/2$ and no larger active coordinate, so its distance from zero is at most $c_* L_{\text{opt}}(\varepsilon)$. Apply [theorem 4.1](#). $\square$

The formulas (4.11) reduce the evaluation of c_* to a maximization over one variable. Numerically, $c_* \approx 1.37486420044$; this value is not proved. The theorem uses only the definition of c_* and the rational bounds, and needs neither uniqueness of the maximizer nor a guaranteed decimal accuracy.

5 Objective distributions and the amount of primal movement

This section takes all scales equal to one. Pool the r positive objective weights and order them as $w_1 \geq \dots \geq w_r > 0$. The objective gap $E(\eta)$ of the central point and the squared central speed in the logarithmic parameter are

$$E(\eta) = \sum_i w_i (1 - q(\eta w_i)), \quad r_{\text{eff}}(\eta) = \sum_i \left(1 - \frac{1}{\sqrt{1 + (\eta w_i)^2}} \right). \quad (5.1)$$

Thus the primal speed depends only on the active channels. For a matrix factor, $r_{\text{eff}} = \text{tr}[I - (I + \eta^2 CC^*)^{-1/2}]$; for products the traces add. Inactive channels contribute zero even if the barrier parameter of the domain is large.

Proposition 5.1 (Gap and path length in terms of the weights). *Let $Q_j(\eta) = \sum_i \min((\eta w_i)^j, 1)$ for $j = 1, 2$, and set $v_0 = \sqrt{1 - 1/\sqrt{2}}$. Then*

$$(2 - \sqrt{2})Q_1(\eta)/\eta \leq E(\eta) \leq Q_1(\eta)/\eta, \quad (5.2)$$

$$v_0 \int_{\eta_0}^{\eta_1} \sqrt{Q_2(\eta)} \frac{d\eta}{\eta} \leq L_{\text{CP}}(\eta_0, \eta_1) \leq \int_{\eta_0}^{\eta_1} \sqrt{Q_2(\eta)} \frac{d\eta}{\eta}. \quad (5.3)$$

Here $\eta > 0$ in (5.2) and $0 \leq \eta_0 < \eta_1$ in (5.3); for $\eta_0 = 0$ the improper integral converges.

Proof. The identity $z(1 - q(z)) = 1 + z - \sqrt{1 + z^2}$ gives

$$(2 - \sqrt{2}) \min(z, 1) \leq z(1 - q(z)) \leq \min(z, 1).$$

For $0 < z \leq 1$, divide by z and use that q increases; for $z \geq 1$ use monotonicity of $1 + z - \sqrt{1 + z^2}$ and its limit one. Furthermore,

$$1 - (1 + z^2)^{-1/2} = \frac{z^2}{\sqrt{1 + z^2}(1 + \sqrt{1 + z^2})} \in [(1 - 1/\sqrt{2}) \min(z^2, 1), \min(z^2, 1)].$$

For $z \leq 1$ bound its displayed denominator between 2 and $2 + \sqrt{2}$; for $z \geq 1$ use monotonicity. Sum and integrate. $\square$

The integral in (5.3) has a closed form between consecutive breakpoints $1/w_i$. If m channels have $\eta w_i \geq 1$ and $B = \sum_{i>m} w_i^2$, the integrand is $\sqrt{m + B\eta^2}/\eta$. For $m, B > 0$ an antiderivative is

$$\sqrt{m + B\eta^2} + \sqrt{m} \log \frac{\sqrt{B}\eta}{\sqrt{m} + \sqrt{m + B\eta^2}}.$$

For $m = 0$ it is $\sqrt{B}\eta$; for $B = 0$ it is $\sqrt{m} \log \eta$. Thus the bounds need only the weights, not a path in the matrix space.

Corollary 5.2 (Central length bounds from the m largest weights). *Assume $w_1 \leq 1$ and $0 < \varepsilon \leq 1$. If $1 \leq m \leq r$ and $\sum_{i>m} w_i \leq \varepsilon/2$, then the central point at $\eta_f = 2m/\varepsilon$ is ε -accurate and*

$$L_{\text{CP}}(\eta_f) \leq \sqrt{(m + \varepsilon^2/4)/2} + \sqrt{2m} \log(2m/\varepsilon). \quad (5.4)$$

Without a tail condition, the central point at r/ε is ε -accurate, with central length at most $\|w\|_2/\sqrt{2} + \sqrt{r} \log(r/\varepsilon)$. More generally, for $\eta_f \geq 1$ and any m , with $\tau_2^2 = \sum_{i>m} w_i^2$,

$$L_{\text{CP}}(\eta_f) \leq \|w\|_2/\sqrt{2} + \sqrt{m + \eta_f^2 \tau_2^2/2} \log \eta_f. \quad (5.5)$$

Proof. The error of the first m channels is at most m/η , and that of the tail is at most its weight sum. For $\eta \leq 1$, the squared speed is at most $\eta^2 \|w\|_2^2/2$, which gives the term $\|w\|_2/\sqrt{2}$. For $1 \leq \eta \leq \eta_f$, squared speed is at most $m + \eta_f^2 \tau_2^2/2$, proving (5.5). For (5.4), a sharper tail estimate uses $\min(z^2, 1) \leq z$: squared speed is at most $m + \eta \sum_{i>m} w_i \leq 2m$. Also $\|w\|_2^2 \leq m + (\sum_{i>m} w_i)^2$. The statement without a tail condition uses $E \leq r/\eta$ and speed at most $\sqrt{r}$. $\square$

For $0 < R < 1$, by lemma 2.1, a central arc of length L can be followed by a forward- R sequence of at most $\lceil L/\log(1+R) \rceil$ steps. These upper bounds concern only primal movement. Consider a conic formulation whose cone barrier restricts to the unit-scale spectral product barrier up to an additive constant, with the same objective and parameter η . Its projected central path is then the path studied here, with primal length measured in the restricted Hessian metric. If its central duality gap is ν/η , substituting $\eta_f = \nu/\varepsilon$ in (5.5) bounds the primal length needed to reach gap ε . For example, if $\tau_2 \leq \varepsilon\sqrt{m}/\nu$, $w_1 \leq 1$, and $0 < \varepsilon \leq \min(1, \nu)$, the primal length is at most $\sqrt{(m + m\varepsilon^2/\nu^2)/2} + \sqrt{3m/2} \log(\nu/\varepsilon)$. This bound excludes the dual movement needed to reach that gap, which section 9 treats.

5.1 Explicit lower bounds and their limits

The exact distance of theorem 3.3 dominates the following determinant-based bound, which shows orders of growth more directly. Let $\bar{w}_{g,m} = (\prod_{i \leq m} w_i)^{1/m}$.

Proposition 5.3 (Determinant-based lower bound). *For every $0 < \varepsilon < \sum_i w_i$,*

$$L_{\text{opt}}(\varepsilon) \geq \max_{1 \leq m \leq r} \sqrt{m} \left[\log \frac{m \bar{w}_{g,m}}{2\varepsilon} \right]_+. \quad (5.6)$$

The bound also holds without the factor 2, as the proof shows from the exact distance formula. Neither form determines L_{opt} up to a universal constant factor.

Proof. For any feasible allocation $x_i \in [0, 1)$, put $e_i = w_i(1 - x_i)$. For a subset I of size m , AM–GM gives $\prod_{i \in I} (1 - x_i) \leq (\varepsilon/m)^m / \prod_{i \in I} w_i$. Since $\rho(x) \geq \log(1/(1 - x))$, Cauchy–Schwarz gives

$$\left(\sum_i \rho(x_i)^2 \right)^{1/2} \geq \sqrt{m} \left[\log \frac{m(\prod_{i \in I} w_i)^{1/m}}{\varepsilon} \right]_+.$$

Minimize over allocations and choose the top m weights. This proves the stronger claim and hence (5.6). To show that the bound with the factor 2 can vanish while $L_{\text{opt}} \rightarrow \infty$, fix $0 < c < \log(2/\sqrt{e})$, $\varepsilon = 1/2$, and set

$$s_i = c/\sqrt{i}, \quad B_r = \sum_{i=1}^r i^{-1/2}, \quad a = \varepsilon/(cB_r), \quad w_i = as_i e^{s_i}.$$

The logarithmic allocation (3.17) has multiplier $1/a$, optimizer s_i , and value $c(\sum_i 1/i)^{1/2} \rightarrow \infty$, so $L_{\text{opt}}(\varepsilon) \rightarrow \infty$ by (3.18). Meanwhile

$$\frac{m\bar{w}_{g,m}}{2\varepsilon} = \frac{m(m!)^{-1/(2m)} e^{(c/m) \sum_{i \leq m} i^{-1/2}}}{2B_r} \leq \frac{e^c \sqrt{e}}{2} \sqrt{m/r} < 1,$$

using $m! \geq (m/e)^m$ and $B_r \geq \sqrt{r}$. Thus every term of (5.6) vanishes although $L_{\text{opt}} \rightarrow \infty$. For the stronger version, keep the same $c < \log(2/\sqrt{e})$. The ratio without 2 is at most $e^c \sqrt{er}/B_r$, uniformly in $m \leq r$. Since $B_r \geq 2(\sqrt{r+1} - 1)$, this upper bound tends to $e^c \sqrt{e}/2 < 1$. Thus even the stronger bound is identically zero for all sufficiently large r . $\square$

The factor 2 comes from a determinant proof, which we give on a matrix ball. Let the columns u_i of U be left singular vectors of the objective for its m largest singular values, with right singular vectors v_i , and set $\psi_U(X) = -\log \det(I - U^* X X^* U)$. Then

$$\nabla^2 \psi_U \preceq \nabla^2 \Phi, \quad |D\psi_U(X)[Z]| \leq \sqrt{m} \|Z\|_{\Phi, X}. \quad (5.7)$$

Indeed, $\Phi = -\log \det \begin{pmatrix} I & X \\ X^* & I \end{pmatrix}$; ψ_U is the negative log determinant of its principal compression. Their difference is the negative log determinant of a Schur complement, which is convex: the Schur complement is matrix concave (the quadratic perspective $B^* A^{-1} B$ is matrix convex), and negative log determinant is convex and order reversing. The second assertion follows from the exact parameter m of the compressed matrix-ball barrier. If $z_i = \text{Re}(u_i^* X v_i)$, Hadamard's inequality and AM–GM give

$$\det(I - U^* X X^* U) \leq \prod_{i \leq m} (1 - z_i^2) \leq 2^m \prod_{i \leq m} (1 - z_i) \leq \frac{(2\varepsilon/m)^m}{\prod_{i \leq m} w_i}.$$

Integrating (5.7) yields (5.6). The allocation proof avoids this compression and also applies to Jordan factors.

5.2 Two decay laws for which the central path has optimal order

Both examples need enough channels for the requested accuracy; for a fixed finite rank, the asymptotics cannot hold as $\varepsilon \downarrow 0$.

For geometric weights $w_i = e^{-a(i-1)}$ with fixed $a > 0$, let $L = \log(1/\varepsilon)$. If L is sufficiently large and $r \geq \lfloor L/a \rfloor$, then

$$L_{\text{opt}}(\varepsilon) = \Theta_a(L^{3/2}), \quad L_{\text{CP}}(\varepsilon) = \Theta_a(L^{3/2}). \quad (5.8)$$

To prove the lower bound, choose $m = \lfloor L/a \rfloor \geq 2$ in (5.6). Its logarithm is $L - a(m-1)/2 + \log(m/2) \geq L/2$, and $m \geq L/(2a)$. For the upper bound take $M = \lceil (L + \log(2/(1 - e^{-a}))) / a \rceil$ and use corollary 5.2 with $\min(M, r)$. The tail is at most $\varepsilon/2$ when $M < r$, and is zero otherwise; $M = O_a(L)$. This gives length $O_a(L^{3/2})$. Thus, for these geometric weights, the ratio $L_{\text{CP}}/L_{\text{opt}}$ stays bounded; geometric decay alone does not make it grow.

For polynomial weights $w_i = i^{-b}$ with fixed $b > 1$, let $m = \lfloor (4\varepsilon)^{-1/(b-1)} \rfloor$. If ε is sufficiently small and $r \geq m$, then

$$L_{\text{opt}}(\varepsilon) = \Theta_b(\varepsilon^{-1/[2(b-1)]}), \quad L_{\text{CP}}(\varepsilon) = \Theta_b(\varepsilon^{-1/[2(b-1)]}). \quad (5.9)$$

For $\eta \geq 1$, splitting at $\lceil \eta^{1/b} \rceil$ and comparing tails with integrals gives $Q_1(\eta) \leq A_b \eta^{1/b}$ and $Q_2(\eta) \leq B_b \eta^{1/b}$, where $A_b = 2 + 1/(b-1)$ and $B_b = 2 + 1/(2b-1)$. Consequently, the central point at $\eta_f = (A_b/\varepsilon)^{b/(b-1)}$ is accurate, and (5.3) bounds its length by $\sqrt{\zeta(2b)/2} + 2b\sqrt{B_b}(\eta_f^{1/(2b)} - 1)$. For the lower bound, $\bar{w}_{g,m} = (m!)^{-b/m} \geq m^{-b}$, so the logarithm in (5.6) is at least $\log 2$. Since $m = \Theta_b(\varepsilon^{-1/(b-1)})$, the orders agree. There is no extra logarithmic factor because new channels activate gradually.

6 A dyadic linear program and step sequences near the central path

We now separate movement near the central path from unrestricted movement on an explicit sparse linear program on the cube, the dyadic family. Except in the last subsection, $r \rightarrow \infty$ and $F(x) = \sum_{i=1}^r b(x_i)$ on $(-1, 1)^r$. We minimize $-w^\top x$, that is, use the support direction $c = w$, so $g_c(x) = \sum_i w_i(1 - x_i)$. Sign changes of coordinates are barrier isometries, so any prescribed signs of the objective give the same results.

Define the following data:

$$h_j = \frac{1}{j \lceil \sqrt{j} \rceil}, \quad A_i = \left\lceil 4r \sum_{j < i} h_j \right\rceil, \quad a_i = A_i \log 2, \quad w_i = 2^{-A_i}, \quad T = a_r, \quad \varepsilon_r = r 2^{-A_r}. \quad (6.1)$$

The box has two inequalities per coordinate, and the constraint matrix has one nonzero per row and two per column. Since $A_i = \Theta(r)$ for every $i \geq 2$, the objective has $\Theta(r^2)$ bits in the ordinary numerator–denominator encoding, and ε_r has $\Theta(r)$ bits. Integer square roots, rational summation, and rounding down compute the data in polynomial time. An exponent encoding of w would be shorter.

Theorem 6.1 (Three primal comparisons on the dyadic family). *Let $x(s)$ denote the central point at $\eta = e^s$. For all sufficiently large r , the first logarithmic parameter s_ε at which the central point is ε_r -accurate satisfies $T - 1 < s_\varepsilon \leq T$, and*

$$L_{\text{opt}}(\varepsilon_r) = \Theta(r\sqrt{\log r}), \quad d_F(0, x(s_\varepsilon)) = \Theta(r\sqrt{\log r}), \quad L_{\text{CP}}(\varepsilon_r) = \Theta(r \log r). \quad (6.2)$$

Consequently, for fixed $0 < R < 1$, the least number of steps of a forward- R sequence from 0 to an ε_r -accurate point is $\Theta_R(r\sqrt{\log r})$. For the standard barriers with equal scales and objectives with r active channels, the supremum of $L_{\text{CP}}(\varepsilon)/L_{\text{opt}}(\varepsilon)$ over objectives and accuracies is $\Theta(\sqrt{\log r})$.

Proof. Write $\bar{a}_i = 4r(\log 2) \sum_{j < i} h_j$ and $e_i = a_i - \bar{a}_i \in (-\log 2, 0]$. The inequalities $\frac{1}{2}j^{-3/2} \leq h_j \leq j^{-3/2}$ show $T = \Theta(r)$ and

$$T - a_i = 4r(\log 2) \sum_{j=i}^{r-1} h_j + O(1), \quad a_{j+1} - a_j \geq 2r(\log 2)j^{-3/2} - \log 2. \quad (6.3)$$

The upper error estimate in [proposition 5.1](#) gives $E(e^T) \leq re^{-T}$. For a sufficiently large absolute constant K , all $i \leq r - K\sqrt{r}$ have $T - a_i \geq 1$: sum the lower bound on h_j over the last $K\sqrt{r}$ terms and use the bounded rounding error. At $s = T - 1$, these channels have $e^{s-a_i} \geq 1$, so $E(e^{T-1}) \geq e^{-(T-1)}(2 - \sqrt{2})(r - O(\sqrt{r})) > re^{-T}$. Since the error decreases strictly, this proves $T - 1 < s_\varepsilon \leq T$.

For $u \geq -1$, [\(4.3\)](#) gives $H(u) = \Theta(1 + u_+)$ with universal constants. Tail summation in [\(6.3\)](#) gives $\sum_i (T - a_i)^2 = \Theta(r^2 \log r)$: the upper bound follows from $T - a_i \leq Cr/\sqrt{i}$, and the lower from $T - a_i \geq cr/\sqrt{i}$ for $i \leq r/16$, after adjusting constants and discarding finitely many small r . With $T - 1 < s_\varepsilon \leq T$, this proves the middle assertion of [\(6.2\)](#) and hence an upper bound on L_{opt} . For the lower bound on L_{opt} , every point x with $g_c(x) \leq \varepsilon_r$ satisfies $w_i(1 - x_i) \leq \varepsilon_r$, hence $\rho(x_i) \geq [T - a_i - \log r]_+$. On $i \leq \sqrt{r}$ these lower bounds are at least $c'r/\sqrt{i}$ for large r , and their squared sum is $\Omega(r^2 \log r)$. Apply [theorem 3.3](#).

Let $J = \lfloor r^{2/3} \rfloor$. On $[a_j, a_{j+1}]$ the first j velocities are at least v_0 . These intervals for $j \leq J$ end before s_ε , since $T - a_{J+1} = \Theta(r^{2/3})$. Their total length is at least $v_0 \sum_{j \leq J} \sqrt{j}(a_{j+1} - a_j)$. Summation by parts bounds the rounding contribution $\sum_{j \leq J} \sqrt{j}(e_{j+1} - e_j)$ in absolute value by $2(\log 2)\sqrt{J}$. The unrounded sum is $4r(\log 2) \sum_{j \leq J} \sqrt{j}h_j = \Theta(r \log r)$. This proves the central lower bound; [theorem 4.1](#) gives its matching upper bound. [Corollary 3.4](#) gives the optimal count, and [theorem 4.3](#) gives the upper bound on the ratio. Scalar intervals embed as diagonal spectral or Jordan channels, so the lower bound on the ratio holds for all these barriers. $\square$

6.1 Step sequences near the central path

We now count the steps of a sequence $z_0, \dots, z_N$ in a metric tube of radius δ_r around the central path: each z_k lies within distance δ_r of a central point $x(s_k)$ with label s_k . Labels may decrease and are not prescribed at either end; instead, z_0 is the analytic center and z_N has objective gap at most ε_r .

Theorem 6.2 (Discrete neighborhood lower bound). *Fix $0 < R < 1$. For the family [\(6.1\)](#), suppose*

$$z_0 = 0, \quad g_c(z_N) \leq \varepsilon_r, \quad \|z_{k+1} - z_k\|_{F, z_k} \leq R, \quad d_F(z_k, x(s_k)) \leq \delta_r \quad (6.4)$$

for arbitrary labels $s_k \in \mathbb{R} \cup \{-\infty\}$, where $x(-\infty) = 0$. If $\delta_r = o(r^{2/3})$ and $C_r = 2\delta_r - \log(1 - R)$, then

$$N = \Omega_R \left(\frac{r \log r}{C_r \sqrt{1 + C_r}} \right). \quad (6.5)$$

In particular, a tube of fixed radius δ gives the lower bound $\Omega_{R, \delta}(r \log r)$, matching exact central-arc sampling. Its ratio to the optimal unrestricted count in [theorem 6.1](#) tends to infinity whenever $\delta_r = o((\log r)^{1/3})$.

Proof. The triangle inequality and [lemma 2.1](#) bound consecutive center distances by C_r . Define

$$J = \lfloor r^{2/3} \rfloor, \quad b = a_{J+1}, \quad n(u) = \#\{i : a_i \leq u\}, \quad P(s) = \int_0^{\min(\max(s, 0), b)} \sqrt{n(u)} \, du.$$

For large r , the first J threshold gaps $a_{j+1} - a_j$ are at least $g = \log 2$ by [\(6.3\)](#). Also $P(b) = \Theta(r \log r)$ by the harmonic sum in the proof of [theorem 6.1](#).

Consider any pair of labels and clip both into $[0, b]$. Because every $H(s - a_i)$ increases, clipping cannot increase the Euclidean distance between their coordinate vectors. Order the clipped labels as $u \leq t$. If $u = t$, P does not change. Otherwise put $j = n(u) \geq 1$, $\Delta = t - u$, and $D = C_r/v_0$.

The first j coordinates have velocity at least v_0 throughout $[u, t]$, so $\Delta \leq D/\sqrt{j}$. Uniform threshold spacing gives $n(t) \leq j + 1 + \Delta/g$ (endpoint values do not affect the integral). Therefore

$$|P(s_{k+1}) - P(s_k)| \leq \Delta \sqrt{j + 1 + \Delta/g} \leq D \sqrt{2 + D/g} = O(C_r \sqrt{1 + C_r}). \quad (6.6)$$

The argument covers jumps across either clipping boundary and to smaller labels.

Next we force the required endpoint progress. The accuracy of z_N and $w_1 = 1$ imply $z_{N,1} \geq 1 - \varepsilon_r$, since each coordinate's objective gap is nonnegative. Scalar metric contraction and the terminal tube give $H(s_N) \geq \rho(1 - \varepsilon_r) - \delta_r \geq T - \log r - \delta_r$. Since $H(s) \leq 1/\sqrt{2} + s_+$, this implies $s_N \geq T - \log r - \delta_r - 1/\sqrt{2} > b$ for large r ; this step uses $T - b = \Theta(r^{2/3})$. To bound the initial progress, we compare P with the length of the central arc: on $[0, b]$ the derivative $\sqrt{n(s)}$ of P is at most $\|(v(s - a_i))\|_2/v_0$. Hence

$$P(s_0) \leq v_0^{-1} L_{CP}(e^{s_0}) \leq v_0^{-1} \Gamma_r d_F(0, x(s_0)) \leq v_0^{-1} \Gamma_r \delta_r = o(r \log r).$$

For $s_0 \leq 0$ the first inequality is immediate because $P(s_0) = 0$; for $s_0 > 0$ it follows by integration. Thus the net progress is $\Theta(r \log r)$. Sum absolute increments in (6.6); no monotonicity hypothesis is used. The final ratio follows from [theorem 6.1](#) because $C_r \sqrt{1 + C_r} = o(\sqrt{\log r})$ in the stated regime. $\square$

The label $-\infty$ covers exact central sequences from the analytic center, even for tube radius zero. The proof also holds with finite labels only; then no radius-zero sequence starts at the analytic center.

Lemma 6.3 (Newton decrement implies a metric tube). *Let $f = F - \langle c, \cdot \rangle$ have an interior minimizer x^* , where F is standard self-concordant. Write $\lambda_f(z) = \|\nabla f(z)\|_{F,z,*}$. If $\lambda_f(z) \leq \beta < 1/2$, with $\beta \geq 0$, then*

$$d_F(z, x^*) \leq \delta_\beta := \log \frac{1 - \beta}{1 - 2\beta}. \quad (6.7)$$

Proof. Put $h = x^* - z$ and $t = \|h\|_{F,z}$. Along the segment, the scalar local norm $a(u) = \|h\|_{F,z+uh}$ satisfies $|a'| \leq a^2$, hence $a(u) \geq t/(1 + ut)$. Since $\nabla f(x^*) = 0$,

$$-Df(z)[h] = \int_0^1 D^2 f(z + uh)[h, h] du \geq \int_0^1 \frac{t^2}{(1 + ut)^2} du = \frac{t^2}{1 + t}.$$

Dual Cauchy–Schwarz gives $t \leq \lambda_f(z)/(1 - \lambda_f(z)) < 1$. The straight segment from z to x^* has length at most $-\log(1 - t)$; substituting the bound on t proves the claim. If $t = 0$, the conclusion holds directly. $\square$

Therefore [theorem 6.2](#) applies to every forward- R sequence that satisfies the usual Newton-decrement condition $\|\nabla F(z_k) - e^{s_k} w\|_{F,z_k,*} \leq \beta < 1/2$ with fixed β and ends at an ε_r -accurate point. It also allows $\beta_r \uparrow 1/2$: the ratio to the optimal count still tends to infinity whenever $\delta_{\beta_r} = o((\log r)^{1/3})$.

The lower bound also applies to feasible conic methods. Suppose a cone barrier $\hat{F}$ restricts on an affine slice $z = z_0 + Bx$ to $F(x)$ up to an additive constant, and suppose $q = c - A^*y$ is a feasible slack for a conic minimization problem, with $AB = 0$ and $B^*c = -w$. The dual norm of the restricted covector $B^*\ell$ in the restricted Hessian is at most the full dual norm of ℓ : both are suprema of $\ell[h]$ over local unit vectors, and restriction reduces that set. Consequently the conventional residual condition

$$\|q + \mu \nabla \hat{F}(z)\|_{\hat{F},z,*} \leq \beta \mu$$

implies the primal decrement bound at parameter $1/\mu$, since $B^*(q + \mu \nabla \widehat{F}(z))/\mu = \nabla F(x) - w/\mu$. A bound on a step in a full primal–dual product norm bounds its primal component and hence the restricted primal chord. By weak duality, a feasible final pair with duality gap at most ε_r gives primal accuracy ε_r . Under these hypotheses the lower bound applies to the conic method. We do not construct a feasible primal–dual sequence that attains the smaller unrestricted count of [theorem 6.1](#).

The same lower bounds hold on a matrix ball or Jordan interval with at least r channels: place the positive weights in a fixed spectral frame and set the other objective weights to zero. Intermediate points need not stay in that frame. The contraction in [theorem 3.1](#) bounds their distance to each labeled center, and the spectral trace inequality turns final accuracy into a lower bound $1 - \varepsilon_r$ on the largest singular value or Jordan eigenvalue. These are exactly the two facts about arbitrary iterates used in the proof above.

6.2 Unequal scales can produce a larger discrete separation

With unequal scales, [\(4.8\)](#) allows the worst-case ratio of central-arc length to endpoint distance to exceed Γ_r . We show a larger separation for step sequences whose labels must cross a prescribed parameter range, unlike in [\(6.4\)](#). In this subsection the barrier is $F_\alpha(x) = \sum_{i=1}^r \alpha_i b(x_i)$ on $(-1, 1)^r$; all local norms, distances, lengths, and labeled centers refer to this weighted barrier. Fix $B \geq 4$ and an integer r . For large U , put

$$u_i = UB^{r-i}, \quad H_0 = H(u_1), \quad \alpha_i = \left(\frac{H_0}{H(u_i)} \right)^2 \geq 1, \quad w_i = \alpha_i e^{u_i}, \quad i = 1, \dots, r.$$

By [\(3.3\)](#), the metric of F_α is Euclidean in the weighted metric coordinates $\sqrt{\alpha_i} \rho(x_i)$. Since $a_i = -u_i$, [\(4.2\)](#) gives the i th of these coordinates at the central point with label s as $\sqrt{\alpha_i} H(s + u_i)$. At the final label $s = 0$, all of them equal H_0 , so the endpoint distance is $H_0 \sqrt{r}$. The central path has length $\Theta(H_0 r)$, uniformly for large U : for fixed $M > 0$, on each of the disjoint label intervals (cores) $I_i = [-u_i + M, -u_{i+1} - M]$, $i < r$, and $I_r = [-u_r + M, 0]$, the i th weighted metric coordinate increases by $\Theta(H_0)$. Indeed, $H(t) = t + O(1)$ for positive t makes this gain at least a fixed fraction of H_0 ; the upper bound follows from [theorem 4.1](#) and $\Gamma \leq \sqrt{r}$.

For fixed $0 < R < 1$ and $\delta \geq 0$, consider forward- R sequences that start at $z_0 = 0$, stay within distance δ of their labeled central points, and have labels $s_0 \leq -u_1 + M$ and $s_N \geq 0$; the initial label may be $-\infty$. Once U is large enough, such a sequence needs $\Omega_{R,\delta}(H_0 r)$ steps. For the proof, take as progress potential the sum over i of the gain of the i th weighted metric coordinate within I_i , with labels clipped to I_i . Every full core has gain exceeding $C = 2\delta - \log(1 - R)$. A permitted center-to-center jump can meet at most two cores in positive length: meeting three would cross the middle one fully, contradicting its distance bound. The at most two clipped coordinate gains sum to at most $\sqrt{2}C$ by Cauchy–Schwarz and exact coordinate distance. Net progress is $\Theta(H_0 r)$, even with backtracking. To compare the same endpoints, specialize to $z_N = x(0)$ and $s_N = 0$. The lower bound still applies, while sampling a shortest route from 0 to $x(0)$ takes $O_R(H_0 \sqrt{r})$ steps. The construction needs large ranges of barrier scales and objective weights; [\(4.8\)](#) shows why such ranges are necessary. We do not claim that its data have polynomial encoding length.

7 Dependence on the barrier

The factor Γ_r of [section 4](#) does not follow from self-concordance alone: its proof uses that the metric velocities of the central coordinates are ordered like the objective weights. A condition on a scalar

barrier guarantees this order; separable barriers built from it keep the sharp factor Γ_r and the dyadic step separation $\Theta(\sqrt{\log r})$. Without the condition and with a growing barrier parameter, the ratio of central-arc length to endpoint distance can approach $\sqrt{r}$. [Section 8](#) shows that the factor Γ_r , and with it the separation $\Theta(\sqrt{\log r})$, also appears for explicit coupled barriers with the least possible parameter on the cube.

7.1 Normalized scalar barriers have the same sharp factor

Let f be a C^3 standard self-concordant barrier on an open interval $I = (\ell, b_f)$, with $f'' > 0$ and an analytic center $x_f \in I$. Assume that f' maps (x_f, b_f) onto $(0, \infty)$ and that

$$(f'(x))^2 \leq f''(x) \quad (x_f < x < b_f). \quad (7.1)$$

We call (7.1) the normalization; it is required only on the positive branch (x_f, b_f) . Define

$$\rho_f(x) = \int_{x_f}^x \sqrt{f''(u)} du, \quad h_f(t) = \rho_f((f')^{-1}(e^t)), \quad v_f = h'_f.$$

The map ρ_f is an isometry from the Hessian metric of f to the line, so distances for the product barrier $F_f(x) = \sum_{i=1}^r f(x_i)$ are Euclidean in the coordinates $\rho_f(x_i)$. Moving a coordinate below x_f up to x_f decreases the distance from the center and improves every objective with positive weights. Coordinates with zero weight stay at x_f ; from now on, r denotes the number of positive weights.

Theorem 7.1 (Normalized scalar comparison). *Under these hypotheses, every central arc for an objective with positive weights satisfies*

$$L_{\text{CP}}(\eta_0, \eta_1) \leq \Gamma_r d_{F_f}(x(\eta_0), x(\eta_1)).$$

For each fixed f and r , the supremum of this ratio for arcs from the analytic center, over positive objectives and terminal parameters, equals Γ_r . Moreover $b_f < \infty$, and for $0 < \varepsilon < (b_f - x_f) \sum_i w_i$,

$$L_{\text{CP}, F_f}(\varepsilon) \leq C_{\text{sc}} \Gamma_r L_{\text{opt}, F_f}(\varepsilon), \quad C_{\text{sc}} < 2. \quad (7.2)$$

Here L_{opt, F_f} is the distance from $(x_f, \dots, x_f)$ to the set where $\sum_i w_i(b_f - x_i) \leq \varepsilon$, and the dimension-independent constant C_{sc} is the explicit maximum attained in [proposition B.1](#). Numerically, $C_{\text{sc}} \approx 1.831856423$.

Proof. At $x = (f')^{-1}(e^t)$, differentiation gives

$$v_f = \frac{f'}{\sqrt{f''}}, \quad v'_f = v_f \left(1 - \frac{v_f f'''}{2(f'')^{3/2}} \right) \geq v_f(1 - v_f) \geq 0. \quad (7.3)$$

Thus v_f is nondecreasing and $0 < v_f \leq 1$. Its limit at $-\infty$ is zero, by continuity at x_f . Its positive limit at $+\infty$ must be one: a smaller limit would make the derivative eventually bounded below by a positive number. The negative tail is integrable because $\int_{-\infty}^t v_f = h_f(t) < \infty$. Once $v_f \geq 1/2$, the differential inequality implies $(1 - v_f)' \leq -(1 - v_f)/2$. Consequently

$$v_f - \mathbf{1}_{(0, \infty)} \in L^1(\mathbb{R}), \quad h_f(t) = t + \kappa_f + o(1) \quad (t \rightarrow \infty). \quad (7.4)$$

Central metric coordinates are $h_f(s + \log w_i)$. Their velocities have the order of the weights, so the prefix proof of [theorem 4.1](#) applies. For sharpness, set $h_f(\log w_i) = M(\sqrt{i} - \sqrt{i-1})$ and terminate at $s = 0$. The integrable step replacement in that proof has error $O_{f,r}(1)$; distance is $M\Gamma_r$ and

length is $M\Gamma_r^2 + O_{f,r}(1)$. This also proves that the supremum equals Γ_r for each fixed scalar barrier f ; f need not vary with the dimension.

Writing $x(t) = (f')^{-1}(e^t)$ gives

$$x'(t) = e^{-t}v_f(t)^2, \quad b_f - x(t) = \int_t^\infty e^{-u}v_f(u)^2 du. \quad (7.5)$$

The first formula makes the right endpoint finite; surjectivity of f' identifies the limit with b_f . The second follows by integration. It remains to prove the accuracy estimate. Put $p_f(y) = f'(\rho_f^{-1}(y))$ on $y \geq 0$. Then

$$p_f(0) = 0, \quad p'_f > 0, \quad |p''_f| \leq p'_f, \quad p_f \leq p'_f. \quad (7.6)$$

A closest accurate metric vector $y^* \geq 0$ exists: intersect the closed feasible set with a ball containing one accurate vector, and use compactness and the finite-coordinate inverse ρ_f^{-1} . The constraint is active, since decreasing a positive coordinate otherwise reduces the norm. Every coordinate is positive. Indeed, increasing a zero coordinate by t improves the gap to first order while increasing squared norm only to second order; compensating by decreasing a positive coordinate preserves the gap and reduces squared norm to first order. The constraint gradient is nonzero. Necessary Lagrange multiplier conditions at this global minimizer therefore give

$$y_i^* p'_f(y_i^*) = \lambda w_i, \quad \lambda > 0. \quad (7.7)$$

For general f , we claim neither convexity of this problem nor uniqueness of its minimizer. With $k = \log 2$, [proposition B.1](#) proves

$$p_f(y) \leq y p'_f(y)/k \leq p_f(C_{\text{sc}} y).$$

The center at $\eta = \lambda/k$ thus has metric coordinates between y_i^* and $C_{\text{sc}} y_i^*$. It is accurate, and the first accurate center is coordinatewise no farther. Apply the endpoint bound. $\square$

More generally, by (7.3), v_f is nondecreasing exactly when $2(f'')^2 - f'f''' \geq 0$ on the positive branch. Monotonicity of v_f alone proves the upper bound Γ_r ; the normalization also provides the limiting speed one, the integrable tails, and (7.2).

Proposition 7.2 (The full self-concordant scalar class). *Consider smooth self-concordant barriers f on an interval that have an analytic center and whose derivative takes every positive value. For the product of r copies of f , the supremum over all such f and all central arcs from the analytic center of the ratio of arc length to endpoint distance is $\sqrt{r}$. In the family that approaches this supremum, the scalar barrier parameter is $\nu_A = \Theta(A^2)$ with $A \rightarrow \infty$. Thus the assertion concerns neither one fixed normalized barrier nor a parameter bounded independently of the dimension.*

Proof. Every central metric coordinate is increasing, whether or not its velocity is monotone. Hence $L \leq \int \|\dot{y}\|_1 = \|y_{\text{end}}\|_1 \leq \sqrt{r} \|y_{\text{end}}\|_2$. [Appendix B.1](#) gives a globally smooth barrier family and disjoint coordinate-velocity windows approaching equality. $\square$

7.2 Step counts on the dyadic linear program for normalized scalar barriers

Corollary 7.3 (The dyadic linear program for a fixed normalized scalar barrier). *Fix f as in [theorem 7.1](#). Use exactly the weights and tolerance (6.1), with objective gap $\sum_i w_i(b_f - x_i)$ and initial point $(x_f, \dots, x_f)$. Then*

$$L_{\text{opt},F_f}(\varepsilon_r) = \Theta_f(r\sqrt{\log r}), \quad L_{\text{CP},F_f}(\varepsilon_r) = \Theta_f(r \log r).$$

Let $0 < R < 1$ and $\delta_r = o(r^{2/3})$. Every sequence with arbitrary labels that satisfies (6.4) in the metric of F_f , with this initial point and objective gap, satisfies (6.5). The constants may depend on f and R . In particular, in a metric tube of fixed radius and in a Newton-decrement neighborhood of fixed radius $0 \leq \beta < 1/2$, the least step count exceeds the unrestricted optimum by a factor $\Theta(\sqrt{\log r})$.

Proof. Choose τ_f so that $v_f(t) \geq 1/2$ for $t \geq \tau_f$. By (7.5), each central coordinate has gap at most e^{-t} , and at least $e^{-t}/4$ when $t \geq \tau_f$. Thus the center at T is accurate and the one at $T - \log 8$ is not, for large r : only $O_f(\sqrt{r})$ terminal coordinates fail the lower-tail condition, by (6.3). From (7.4), $|h_f(t) - t_+| \leq B_f$ for all t . Therefore the accurate-center distance is $\Theta_f(r\sqrt{\log r})$. Every accurate endpoint has $b_f - x_i \leq \varepsilon_r/w_i$, which implies

$$\rho_f(x_i) \geq T - a_i - \log r - O_f(1)$$

whenever the right side is large. To see this implication directly, write $x_i = x(t)$ near b_f and combine $b_f - x(t) \geq e^{-t}/4$ with $h_f(t) = t + O_f(1)$. The first $\lfloor \sqrt{r} \rfloor$ coordinates give the matching lower bound on the distance to the accurate set, as in theorem 6.1.

Use the shifted thresholds $a_i + \tau_f$ and cutoff $a_{J+1} + \tau_f$, where $J = \lfloor r^{2/3} \rfloor$. The first j velocities exceed $1/2$ after threshold j . The gap spacing and harmonic sum in (6.3) are unchanged by this common shift. They give central length $\Omega_f(r \log r)$ before the first accurate center; the prefix bound gives the matching upper bound. For discrete sequences define the clipped potential P_f by replacing $0, b$ in the proof of theorem 6.2 with $\tau_f, a_{J+1} + \tau_f$. The exact product metric and velocity lower bound give each progress increment $O_f(C_r \sqrt{1 + C_r})$, by the same calculation as (6.6), with v_0 replaced by $1/2$. The start tube gives $P_f(s_0) \leq 2\Gamma_r \delta_r$ by the prefix bound. Actual accuracy in coordinate one forces $s_N \geq T - \log r - \delta_r - O_f(1) > a_{J+1} + \tau_f$. Hence net progress is $\Theta(r \log r)$ even with backtracking labels. Finally the straight segment in ρ_f coordinates to the center at T has length $O_f(r\sqrt{\log r})$; chord sampling of this segment gives the unrestricted upper count. Lemma 6.3 applies to the product barrier and proves the decrement assertion. $\square$

The dyadic data keep their encoding lengths, but a general fixed f need not be efficiently computable in finite arithmetic; the corollary concerns only movement in the barrier metric.

7.3 Universal and entropic cube barriers

The universal barrier of Nesterov and Nemirovski and the entropic barrier of Bubeck and Eldan [5] are defined on every convex body. Lee and Yue [15] and Chewi [7] proved, respectively, that their parameters are at most the dimension. On the cube, both satisfy the hypotheses of theorem 7.1.

Proposition 7.4 (Canonical barriers on a cube). *For $K = (-1, 1)^r$, the universal barrier is $U(x) = \sum_i b(x_i)$ up to an additive constant. The entropic barrier is $E(x) = \sum_i e(x_i)$, where*

$$e = a^*, \quad a(\theta) = \log \frac{2 \sinh \theta}{\theta}, \quad a(0) = \log 2.$$

Both have parameter exactly r . For each, the worst-case ratio of central-arc length to endpoint distance is exactly Γ_r , the worst-case ratio $L_{\text{CP}}(\varepsilon)/L_{\text{opt}}(\varepsilon)$ is of order $\sqrt{\log r}$, and corollary 7.3 applies.

Proof. In orthant $\sigma \in \{-1, 1\}^r$, the polar of $K - x$ is the simplex $\sum_i (1 - \sigma_i x_i) |z_i| \leq 1$, of volume $[r! \prod_i (1 - \sigma_i x_i)]^{-1}$. Summing gives

$$\text{vol}((K - x)^\circ) = \frac{1}{r!} \prod_i \left(\frac{1}{1 - x_i} + \frac{1}{1 + x_i} \right) = \frac{2^r}{r!} \prod_i (1 - x_i^2)^{-1}.$$

Its logarithm is the claimed universal barrier. Fubini's theorem gives $\log \int_K e^{\langle \theta, z \rangle} dz = \sum_i a(\theta_i)$; separating the supremum in Fenchel conjugacy gives the entropic formula. These product formulas are elementary special cases, not new barrier constructions; the product structure is also explicit in Chewi's tensorization argument [7, Section 4]. The explicit cube log-partition and its coordinatewise gradient also appear in recent work on linear bandits [16, Section 6.1].

For the entropic summand, $a'' > 0$ and

$$x = a'(\theta) = \coth \theta - 1/\theta, \quad a''(\theta) = 1/\theta^2 - 1/\sinh^2 \theta,$$

with their continuous values at zero. The map a' increases from -1 to 1 , and $e'(x) = \theta$. Chewi's theorem gives standard self-concordance with parameter at most one in dimension one. Directly, the squared local gradient norm is

$$\frac{e'(x)^2}{e''(x)} = \theta^2 a''(\theta) = 1 - (\theta/\sinh \theta)^2 \longrightarrow 1 \quad (\theta \rightarrow \infty).$$

Thus the scalar parameter is exactly one and its positive branch satisfies (7.1); its center is zero. The product parameter is exactly r by taking all coordinates to the positive endpoint. The same facts for b were proved in proposition 3.2. Apply the normalized scalar theorem and its dyadic corollary. $\square$

Other interval barriers and their Hessian geometry have been studied. Nesterov and Todd [21, Section 6.2] use $-\log \cos$ on a rescaled interval, while Papa Quiroz and Oliveira [23] construct a different cube barrier with parameter $3r/2$ and study geometric algorithms. Neither the coordinates ρ_f nor other cube barriers are new; proposition 7.4 only states the sharp central-arc comparison and the step-sequence separation for the universal and entropic barriers.

7.4 Extension to trace-separable spectral barriers

The scalar results extend to spectral barriers without assuming that commuting paths (paths in one eigenframe) are globally shortest; this follows from the Hessian formula. If $F(x) = \text{tr } f(x) = \sum_i f(\lambda_i(x))$ on a Jordan spectral interval, its Peirce Hessian is

$$D^2 F(x)[u, u] = \sum_i f''(\lambda_i) u_i^2 + \sum_{i < j} \frac{f'(\lambda_i) - f'(\lambda_j)}{\lambda_i - \lambda_j} \|u_{ij}\|^2. \quad (7.8)$$

At coincident eigenvalues the divided difference is its continuous extension $f''(\lambda_i)$; the Peirce components are normalized as in (3.13). One can obtain (7.8) directly: polynomial Jordan calculus gives $D \text{tr } f(x)[u] = \langle f'(x), u \rangle$ and makes $Df'(x)$ act on V_{ij} by the displayed divided difference. Approximate f'' uniformly by polynomials on a compact interval containing the spectrum and integrate twice. The resulting polynomial functions and their first two derivatives converge uniformly; their divided differences do also, by their integral representation. This proves the formula for C^2 functions. It is the trace-separable case of the classical spectral differential calculus [4, 17].

The off-diagonal Peirce terms are nonnegative, since f' is increasing. The almost-everywhere spectral derivative argument in theorem 3.1 consequently gives

$$\|\dot{x}\|_{F,x}^2 \geq \sum_i f''(\lambda_i) \dot{\lambda}_i^2 = \left\| \frac{d}{dt} \rho_f(\lambda(x)) \right\|^2.$$

Thus distances from the center, and distances between points with a common frame and equally ordered eigenvalues, are Euclidean distances between their spectra after applying ρ_f . For a support

objective with positive eigenvalues w_i , the central points lie in the objective frame and satisfy $f'(\lambda_i) = \eta w_i$. Hence [theorem 7.1](#) applies, with r equal to the rank of the objective. For the accuracy bound, the Jordan trace inequality aligns an endpoint with the objective frame, in the same order; replacing eigenvalues below x_f by x_f improves both the objective and the distance. The scalar minimizer and necessary conditions [\(7.7\)](#) then give the same bound.

For rectangular matrices, assume f is even on $(-b_f, b_f)$, $x_f = 0$, and $f'' > 0$. Apply the Hermitian dilation argument with $g = f/2$ to $F(X) = \sum_i f(\sigma_i(X))$; the two copies of each singular value recover exactly f'' . The same conclusions hold. These statements concern only Hessian metrics; we do not claim that the spectral extension of a self-concordant scalar barrier is standard self-concordant. The corollaries about bounded steps need this property for the spectral barrier; [proposition 3.2](#) proves it for the standard spectral barriers with all scales at least one.

8 Coupled cube barriers with optimal parameter

The barriers of [section 7](#) are separable. We show that the worst-case ratios, between endpoints and at equal accuracy, stay at least Γ_r for some coupling terms that make the Hessian non-diagonal: we prove this for convex couplings bounded near one facet, give two coupled barriers with the optimal parameter, and prove a matching upper bound for one of them. Throughout, $K = (-1, 1)^r$ and $U(x) = \sum_i b(x_i)$. Every standard self-concordant barrier on K has parameter at least r ; this classical bound follows because the vertices of K are simple [\[20, Proposition 2.3.6\]](#).

8.1 A lower bound when the coupling is bounded near a facet

For $\delta \in (0, 1)$, the facet collar of width δ is $\{x \in K : x_1 \geq 1 - \delta\}$.

Theorem 8.1 (Lower bound under bounded coupling near a facet). *Let $F = U + G$ be a standard self-concordant barrier, with $G \in C^2(K)$ convex. Suppose that, after a signed coordinate permutation, some $\delta \in (0, 1)$ and finite M, C satisfy*

$$\|\nabla G(x)\|_\infty \leq M, \quad 0 \leq G''(x) \preceq CI \quad \text{for every } x \in K \text{ with } x_1 \geq 1 - \delta. \quad (8.1)$$

Let x_ be the analytic center of F . Then*

$$\sup_{w>0, \eta>0} \frac{L_{CP,F}(0, \eta)}{d_F(x_*, x_w(\eta))} \geq \Gamma_r, \quad \sup_{w>0, 0<\varepsilon<g_w(x_*)} \frac{L_{CP,F}(\varepsilon)}{L_{opt,F}(\varepsilon)} \geq \Gamma_r. \quad (8.2)$$

In the second ratio, distances to accuracy are measured from x_ . The weights are positive in the permuted coordinates. We do not claim a matching upper bound for all G satisfying [\(8.1\)](#).*

Proof. The bounded cube and barrier blow-up give an interior minimizer, unique because $F'' \succeq U'' \succ 0$. Let $d_i = \sqrt{i} - \sqrt{i-1}$, choose $w_i = e^{Md_i}$ with $M \rightarrow \infty$, and write $x_M(s)$ for the center at $\eta = e^s$, terminating at $s = 0$. In this proof only, denote the gradient bound in [\(8.1\)](#) by M_G to distinguish it from M . Set $z_i = e^{s+Md_i}$. For each fixed x_{-1} , convexity implies

$$x_1 \leq 1 - \delta \implies \partial_1 G(x_1, x_{-1}) \leq \partial_1 G(1 - \delta, x_{-1}) \leq M_G.$$

Choose $Z > M_G + b'(1 - \delta)$. The equation $b'(x_1) + \partial_1 G = z_1$ shows that $z_1 \geq Z$ forces entry into the collar. As z_1 increases, the path stays there.

In the collar, put $D = U''(x_M)$ and $H = F''(x_M)$. Differentiated stationarity gives $H\dot{x}_M = z$ and speed squared $z^T H^{-1} z$. For any coordinate subset A , the variational formula for this inverse quadratic form, restricted to vectors supported on A , gives

$$z^T H^{-1} z \geq z_A^T H_{AA}^{-1} z_A \geq \sum_{i \in A} \frac{z_i^2}{b''(x_{M,i}) + C}. \quad (8.3)$$

Also $b'(x_{M,i}) = z_i - \partial_i G$ with $|\partial_i G| \leq M_G$. Because $(b')^2/b'' = 2x^2/(1+x^2) \rightarrow 1$ as $x \uparrow 1$, increasing Z makes every summand in (8.3) at least $1 - \zeta$ whenever $z_i \geq Z$, for any fixed $\zeta > 0$. The activation times $\tau_i = \log Z - Md_i$ are strictly ordered, and all lie below zero for large M . Integrating over their successive prefix intervals gives

$$L_{\text{CP},F}(0,1) \geq \sqrt{1-\zeta} [M\Gamma_r^2 - \sqrt{r} \log Z]. \quad (8.4)$$

At termination, stationarity and the bounded gradient imply $y_i := \rho(x_{M,i}(0)) = Md_i + O_{r,M_G}(1)$, by (4.3); hence $\|y\| = M\Gamma_r + O(1)$.

Fix an interior point p with $p_1 = 1 - \delta/2$ and a fixed compact interior curve from x_* to p . Its F -length is finite. The straight segment from $\rho(p)$ to y in product metric coordinates has monotone physical coordinates. For large M its first coordinate stays above $1 - \delta$, so it lies in the collar. Its U -length is $M\Gamma_r + O(1)$ and its Euclidean total variation is at most $2r$. Its extra G -length is therefore at most $2r\sqrt{C}$. Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ gives

$$d_F(x_*, x_M(0)) \leq M\Gamma_r + O_{F,r,M_G,C}(1). \quad (8.5)$$

Divide (8.4) by this bound and let first $M \rightarrow \infty$, then $\zeta \downarrow 0$. This proves the first assertion, with analytic-center initialization interpreted as the limit $s \rightarrow -\infty$. Near that limit the path has finite length because $x(\eta) = x_* + \eta F''(x_*)^{-1} w + o(\eta)$.

For each such objective and terminal center, choose $\varepsilon = g_w(x_M(0)) > 0$. The identity

$$\frac{d}{d\eta} w^T x_w(\eta) = w^T F''(x_w)^{-1} w > 0$$

shows this is the first accurate center and $\varepsilon < g_w(x_*)$. Since $L_{\text{opt},F}(\varepsilon) \leq d_F(x_*, x_M(0))$, the same construction proves the second assertion. $\square$

The proof needs the bounds in (8.1) on the whole collar, not only near a vertex: it uses them after the first coordinate enters the collar, while the others are still far from their facets. Hence, for convex G , a worst-case ratio bounded independently of r requires (8.1) to fail for every signed permutation and every δ , that is, derivatives of G unbounded in every facet collar. This is a necessary condition; we do not claim that such a G exists.

8.2 Two explicit coupled barriers with the optimal parameter

Theorem 8.2 (Dense and radial coupling at the cube parameter). *For $r \geq 2$, each of the barriers*

$$F_d(x) = U(x) + \frac{1}{8r} (\mathbf{1}^T x)^2, \quad (8.6)$$

$$F_{\lambda,c}(x) = U(x) - \lambda \log(c - \|x\|^2), \quad \lambda \geq 1, \quad c - r \geq 4\lambda, \quad (8.7)$$

is standard self-concordant with exact parameter r and has nonzero mixed Hessian entries on an open subset of the cube. The dense barrier is invariant under permutations and simultaneous sign reversal. The radial family is invariant under every signed permutation. For each fixed choice of their parameters, both suprema in (8.2) are at least Γ_r .

Proof. Write $p = \nabla U$ and $D = U''$. For the dense example put $\beta = 1/(4r)$, $t = \mathbf{1}^T x$, $A = \mathbf{1}^T D^{-1} \mathbf{1}$, $B = \mathbf{1}^T D^{-1} p$, and $Q = p^T D^{-1} p$. Sherman–Morrison gives

$$\|\nabla F_d\|_{F_d, x, *}^2 = Q + \frac{\beta(2tB - B^2) + \beta^2 t^2 A}{1 + \beta A}.$$

Let $m = r - Q = \sum_i (1 - x_i^2)/(1 + x_i^2) > 0$. The explicit interval derivatives give $|B| \leq m$, $A \leq m/2$, and $|t| < r$. Dropping $-\beta B^2$ bounds the correction by $(2\beta r + \beta^2 r^2/2)m = 17m/32 < m$. Thus the parameter is at most r . The quadratic has zero third derivative and positive semidefinite Hessian, so it preserves standard self-concordance; U supplies boundary blow-up.

For the radial case put $S = c - \|x\|^2$ and $a = 2\lambda/S$. Then $0 < a < 1/2$ and

$$\nabla F_{\lambda, c} = p + ax, \quad F''_{\lambda, c} = D + aI + (a^2/\lambda)xx^T. \quad (8.8)$$

The ball term is an affine rescaling of the rank-one spectral-ball barrier in [proposition 3.2](#); multiplying by $\lambda \geq 1$, restricting to the cube, and summing preserve standard self-concordance. Inverse Loewner order gives

$$\|\nabla F_{\lambda, c}\|_{F_{\lambda, c}, x, *}^2 \leq \sum_i \frac{(p_i + ax_i)^2}{D_i + a} \leq r.$$

For the last inequality set $y = x_i^2$. After multiplying by $(1-y)^2$, the difference between denominator and numerator in its i th summand is

$$(1-y)[2 + a(1-5y) - a^2 y(1-y)].$$

The bracket decreases on $[0, 1]$, since its derivative is $a(2ay - a - 5) < 0$, and its value at one is $2 - 4a \geq 0$.

For both examples, along $x = t\mathbf{1}$ as $t \uparrow 1$ the added gradient and Hessian stay bounded, while the interval gradient and Hessian diverge. The ratio of the squared first to the second directional derivative along $\mathbf{1}$ tends to r ; explicitly it is $r(p_0 + at)^2/[D_0 + a + (a^2/\lambda)rt^2]$ for the radial example. Thus both exact parameters equal r . The stated symmetries follow from the formulas. The mixed Hessians are β in the dense case and $(a^2/\lambda)x_i x_j$ in the radial case. Finally the dense coupling obeys $\|\nabla G\|_\infty \leq 1/4$, $G'' \preceq I/4$. The radial coupling obeys

$$\|\nabla G\|_\infty < 1/2, \quad 0 \preceq G'' \preceq (1/2 + r/(4\lambda))I.$$

These global bounds imply [\(8.1\)](#); apply [theorem 8.1](#). $\square$

Adding the parameters of the two terms in [\(8.7\)](#) gives only $r + \lambda$, whereas the exact parameter is r ; with $c = r + 4\lambda$ and large λ the gap is arbitrarily large. This requires c to vary, since for fixed r and c our sufficient condition allows only $\lambda \leq (c - r)/4$. Quadratic regularization that preserves the parameter of the box barrier is known: Castro and Cuesta [\[6, Section 4.1\]](#) prove this for suitably bounded diagonal regularizations. We use our two examples only for the central-path comparisons and do not claim that parameter-preserving regularization is new.

8.3 A uniform matching upper bound for the radial family

Theorem 8.3 (Upper bound for the radial family). *For every barrier [\(8.7\)](#) and every nonzero linear objective with k nonzero coordinates, every central arc satisfies*

$$L_{\text{CP}, F}(\eta_0, \eta_1) \leq B_r \Gamma_k d_F(x(\eta_0), x(\eta_1)), \quad B_r = \sqrt{\frac{11}{8}} \frac{25}{14}. \quad (8.9)$$

For the central path from the analytic center 0 and $0 < \varepsilon < \sum_i |w_i|$,

$$L_{\text{CP},F}(\varepsilon) \leq \frac{5}{4} B_{\text{rc}} \Gamma_k L_{\text{opt},F}(\varepsilon). \quad (8.10)$$

The constants do not depend on r , λ , or c ; we do not claim that they are sharp. Together with [theorem 8.1](#), these inequalities show that, for every barrier in this family, both worst-case ratios over positive objectives are $\Theta(\sqrt{\log r})$.

Proof. Let $t = \sum_i (1 - x_i^2)$, so $S \geq 4\lambda + t$. The interval Hessian satisfies $D \succeq 2I$ and $x^T D^{-1} x \leq t/2$. The two coupling terms in (8.8) satisfy

$$aI \preceq \frac{1}{4}D, \quad \frac{a^2}{\lambda} x x^T \preceq \frac{2\lambda t}{(4\lambda + t)^2} D \preceq \frac{1}{8}D.$$

The last inequality follows from $(t - 4\lambda)^2 \geq 0$. Consequently, throughout the whole cube,

$$U'' \preceq F'' \preceq \frac{11}{8} U''. \quad (8.11)$$

In particular $d_F \geq d_U$, even for paths that leave the subspace of active (nonzero-weight) coordinates.

By signed-permutation invariance take the nonzero weights positive and decreasing. The stationary equations at $\eta = e^s$ are

$$z_i = e^s w_i = x_i(b_i + a), \quad b_i = 2/(1 - x_i^2). \quad (8.12)$$

Zero-weight coordinates are zero. Every other coordinate is positive, and their order is that of w_i , since $x \mapsto x(b(x) + a)$ is strictly increasing for fixed a . Differentiating with respect to s and solving for a' gives

$$(D_i + a)x'_i + x_i a' = z_i, \quad a' = \frac{(a^2/\lambda)K}{1 + (a^2/\lambda)L}, \quad K = \sum_i \frac{x_i^2(b_i + a)}{D_i + a}, \quad L = \sum_i \frac{x_i^2}{D_i + a}. \quad (8.13)$$

For $y = x_i^2$, the inequality $y(b_i + a)/(D_i + a) \leq 1 - y$ reduces, after multiplication by $1 - y$, to $2 + a(1 - 2y)(1 - y) > 0$. Thus $K \leq t$ and

$$0 \leq a' \leq \frac{4\lambda t}{(4\lambda + t)^2} \leq \frac{1}{4}. \quad (8.14)$$

Put $q_i = b'(x_i) = b_i x_i$ on active coordinates and define

$$\theta_i = (\log q_i)' = \frac{D_i(b_i + a - a')}{(D_i + a)b_i}, \quad v_i = \frac{q_i}{\sqrt{D_i}} = \frac{\sqrt{2}x_i}{\sqrt{1 + x_i^2}}.$$

Since $D_i, b_i \geq 2$, $a \leq 1/2$, and $a' \leq 1/4$,

$$\frac{7}{10} = \frac{4}{5} \frac{7}{8} \leq \theta_i \leq \frac{5}{4}, \quad y'_i := (\rho(x_i))' = \theta_i v_i. \quad (8.15)$$

In particular all coordinates increase. The v_i are nonnegative and decreasing in index, since the x_i are ordered. Let $d_i = \sqrt{i} - \sqrt{i-1}$. Combining the metric comparison, the bounds on θ_i , and the prefix inequality yields

$$\begin{aligned} L_{\text{CP},F} &\leq \sqrt{11/8} \int \|y'\|_2 \, ds \leq \sqrt{11/8} \frac{5}{4} \int \|v\|_2 \, ds \\ &\leq \sqrt{11/8} \frac{5}{4} \sum_{i=1}^k d_i \int v_i \, ds \leq B_r \sum_{i=1}^k d_i \Delta y_i \leq B_r \Gamma_k \|\Delta y\|_2. \end{aligned}$$

The last norm equals the full-cube U -distance of the two centers and is at most their F -distance. This proves (8.9), including arcs that start at the analytic center.

For accuracy, compare with the standard U center for the same weights. Equation (8.12) gives

$$\frac{4}{5} z_i \leq q_i \leq z_i. \quad (8.16)$$

Let η_U be the first accurate U parameter. The radial center at $\bar{\eta} = 5\eta_U/4$ is coordinatewise at least this U center and is therefore accurate. By coordinate monotonicity the first accurate radial center occurs no later. For the function H in (4.1), $v = H'$ has $v' \leq v$ because $b''' \geq 0$ on $(0, 1)$. Integration from $-\infty$ gives $v \leq H$, and hence $H(s + \log a) \leq aH(s)$ for $a \geq 1$. The upper bound in (8.16) consequently shows that the radial ρ coordinates at $\bar{\eta}$ are at most $5/4$ times the coordinates of the first accurate U center. The proof of theorem 4.3 bounds the latter vector's norm by $c_* L_{\text{opt},U}(\varepsilon)$. Finally $L_{\text{opt},U}(\varepsilon) \leq L_{\text{opt},F}(\varepsilon)$ follows from global metric domination. Apply the direct bound $L_{\text{CP},F}(0, \bar{\eta}) \leq B_r \Gamma_k \|y(\bar{\eta})\|$ proved above, obtaining (8.10). $\square$

8.4 The dyadic separation for the radial family

Corollary 8.4 (Uniform radial discrete comparison). *Use the same dyadic objective and tolerance (6.1) with any $F_{\lambda,c}$ satisfying (8.7); the parameters may vary with r . Uniformly over this family,*

$$L_{\text{opt},F}(\varepsilon_r) = \Theta(r\sqrt{\log r}), \quad L_{\text{CP},F}(\varepsilon_r) = \Theta(r \log r).$$

Let $0 < R < 1$ and $\delta_r = o(r^{2/3})$. Every forward- R sequence with arbitrary labels that starts at zero, ends at a point with gap at most ε_r , and stays within F -distance δ_r of its labeled central points satisfies (6.5). An explicit unrestricted sequence reaches accuracy ε_r in $O_R(r\sqrt{\log r})$ steps. In particular, in metric tubes of fixed radius and in Newton-decrement neighborhoods of fixed radius $0 \leq \beta < 1/2$, the least step count exceeds the unrestricted optimum by a factor $\Theta(\sqrt{\log r})$.

Proof. Choose an absolute τ so that $v(\tau + \log(4/5)) \geq 1/2$, with $v = H'$ from (4.1). Equations (8.15) and (8.16) imply

$$y'_i(s) \geq \kappa := 7/20 \quad (s \geq a_i + \tau), \quad 0 \leq y_i(s) \leq H(s - a_i). \quad (8.17)$$

The U center at T is accurate, so the radial center at $T + \log(5/4)$ is accurate by (8.16). The Euclidean norm of its ρ coordinates is $O(r\sqrt{\log r})$ by (6.3). The straight segment from zero to this endpoint in ρ coordinates has F -length at most $\sqrt{11/8}$ times its U -length, by (8.11). Chord sampling of this segment gives the explicit unrestricted sequence. Global metric domination and the lower bound on $L_{\text{opt}}(\varepsilon_r)$ in theorem 6.1 give the matching $\Omega(r\sqrt{\log r})$ distance.

For the lower bound, use thresholds $\sigma_i = a_i + \tau$, $J = \lfloor r^{2/3} \rfloor$, cutoff $b = \sigma_{J+1}$, and the potential

$$P(s) = \int_{\sigma_1}^{\min(\max(s, \sigma_1), b)} \sqrt{\#\{i : \sigma_i \leq u\}} \, du.$$

Its total range is $\Theta(r \log r)$, by the same harmonic sum and bounded rounding error as in [theorem 6.2](#). The step bound R and the two tube conditions bound the F -distance between consecutive labeled centers by $C_r = 2\delta_r - \log(1 - R)$. By metric domination this also bounds the Euclidean distance between their y coordinates. All coordinates increase by (8.15), so clipping labels does not increase this distance. The first j active coordinates advance at rate at least κ , and successive thresholds are separated by at least $\log 2$. Thus the calculation (6.6), with v_0 replaced by κ , bounds a progress jump by $O(C_r \sqrt{1 + C_r})$.

Initially, the direct prefix estimate in the proof of [theorem 8.3](#) and (8.17) give

$$P(s_0) \leq \kappa^{-1} \int_{-\infty}^{s_0} \|y'(s)\|_2 ds \leq \kappa^{-1} \frac{25}{14} \Gamma_r \|y(s_0)\|_2 \leq \kappa^{-1} \frac{25}{14} \Gamma_r \delta_r.$$

At termination $1 - z_{N,1} \leq \varepsilon_r$; scalar contraction and the terminal tube imply $y_1(s_N) \geq T - \log r - \delta_r$. Together with $y_1(s) \leq H(s) \leq s_+ + 1/\sqrt{2}$, this forces $s_N > b$ for large r . Since $T - b = \Theta(r^{2/3})$, the assumption $\delta_r = o(r^{2/3})$ suffices here. The net progress is $\Theta(r \log r)$; summing the jump bound over all steps proves (6.5) without assuming monotone labels. Applying the same endpoint argument to the first accurate center and integrating (8.17) gives the central-length lower bound; [theorem 8.3](#) gives its upper bound. Finally apply [lemma 6.3](#). $\square$

Example 8.5 (A coupling singular at the vertices). For $r \geq 2$, consider $F_0 = U - \log(r - \|x\|^2)$. It has exact parameter $r + 1$, so it lies outside (8.7). Adding the parameters of the two terms gives the upper bound; along $x = (1 - \tau)\mathbf{1}$ its restriction is $-(r + 1) \log \tau + O(1)$ with the corresponding first two derivative asymptotics, proving the reverse bound. Its endpoint and equal-accuracy suprema are still at least Γ_{r-1} . To prove this despite the singular coupling, set $n = r - 1$, $w_i = e^{Md_i}$ for $i \leq n$, and $w_r = e^{-M^2}$, terminating at $s = 0$. Stationarity gives $0 \leq x_r(s) \leq e^{-M^2}/2$ for $s \leq 0$. Thus $r - \|x\|^2 \geq 1 - x_r^2 \geq 1/2$ on this central arc and on its coordinatewise U -geodesic from zero. The extra gradient is bounded by 4 and its Hessian by $(4 + 16r)I$ on these paths. Apply (8.3) to the first n coordinates. Their prefix intervals give length $M\Gamma_n^2 - o_r(M)$; their endpoint metric coordinates are $Md_i + O_r(1)$, while the final coordinate is $o(1)$. The comparison geodesic has F_0 -length at most $M\Gamma_n + O_r(1)$, since its Euclidean total variation is at most r . The ratio lower bound follows, and choosing the actual terminal gap gives the lower bound on the equal-accuracy supremum exactly as in [theorem 8.1](#). Thus the lower bound does not need (8.1) on a whole facet collar: a coordinate with negligible weight keeps the coupling regular along the central arc and the comparison path.

9 Primal movement and the cost of a feasible dual solution

A short primal route to an accurate point need not extend to a short route ending at a feasible primal–dual pair. We restate the classical primal–dual geometry with its metric and endpoints explicit, then compare the two on the dyadic linear program of [section 6](#).

9.1 Speed allocation and the set of small-gap pairs

Let K be a proper cone (closed, pointed, full dimensional), and let F be a standard self-concordant barrier satisfying $F(tx) = F(x) - \nu \log t$ for $t > 0$. Use the negative-pairing conjugate

$$F_*(s) = \sup_{x \in \text{int } K} \{-\langle s, x \rangle - F(x)\}, \quad \hat{F}(x, s) = F(x) + F_*(s).$$

Consider the strictly feasible primal–dual pair

$$\min\{\langle c, x \rangle : Ax = b, x \in K\}, \quad \max\{\langle b, y \rangle : A^*y + s = c, s \in K^*\},$$

and its exact centers $z(\eta) = (x(\eta), s(\eta))$, where $s = -\eta^{-1}F'(x)$ and $\langle x, s \rangle = \nu/\eta$. Full primal–dual distances use the Hessian metric of $\hat{F}$ on the product cone, ignoring the affine constraints; distances inside the feasible set can only be larger. A dot denotes differentiation with respect to $\log \eta$.

Proposition 9.1 (Classical speed allocation). *Put $H = F''(x)$, $g = H^{-1/2}F'(x)$, and let Π_x be the Euclidean orthogonal projector onto $H^{1/2} \ker A$. Then*

$$\begin{aligned} \nu_P(\eta) &:= \|\dot{x}\|_{F,x}^2 = \|\Pi_x g\|^2, & \nu_D(\eta) &:= \|\dot{s}\|_{F^*,s}^2 = \|(I - \Pi_x)g\|^2, \\ \nu_P(\eta) + \nu_D(\eta) &= \nu. \end{aligned} \tag{9.1}$$

If A has full row rank, then

$$\nu_P(\eta) = \eta^2 \langle c, Mc \rangle, \quad M = H^{-1} - H^{-1}A^*(AH^{-1}A^*)^{-1}AH^{-1}. \tag{9.2}$$

Writing $p(\eta) = \langle c, x(\eta) \rangle$ and $d(\eta) = \langle b, y(\eta) \rangle$, one has

$$-\eta^2 p'(\eta) = \nu_P(\eta), \quad \eta^2 d'(\eta) = \nu_D(\eta). \tag{9.3}$$

If both objectives converge to the finite optimum p^* , then

$$p(\eta) - p^* = \int_{\eta}^{\infty} \frac{\nu_P(u)}{u^2} du, \quad p^* - d(\eta) = \int_{\eta}^{\infty} \frac{\nu_D(u)}{u^2} du. \tag{9.4}$$

Proof. Logarithmic homogeneity and conjugacy give $Hx = -F'(x)$, $\|g\|^2 = \nu$, and $F''_*(s) = \eta^2 H^{-1}$. Differentiating the central relation yields $\eta \dot{s} = F'(x) - H\dot{x}$. Thus $v = H^{1/2}\dot{x}$ and $d_0 = \eta H^{-1/2}\dot{s}$ satisfy $g = v + d_0$. Feasibility gives $v \in H^{1/2} \ker A$ and $d_0 \in (H^{1/2} \ker A)^{\perp}$, proving (9.1). Removing redundant equalities leaves the feasible tangent spaces unchanged. With full row rank, differentiated primal stationarity gives $\dot{x} = -\eta M c$; $MHM = M$ proves (9.2) and the first identity in (9.3). Differentiate $p - d = \nu/\eta$ to obtain the second, and integrate for (9.4). $\square$

The constant total speed is the calculation behind the primal–dual theorem of Nesterov and Todd [21, Theorems 4.1 and 5.2]. Metric projections and Hessian isometries between conjugate barriers also appear in Ohara, Ishi, and Tsuchiya [22, Sections 4–5]. These formulas are not new; we use them to determine exactly how much of the total speed belongs to the primal path.

Theorem 9.2 (Classical distance to feasible small-gap points). *Let $z_0 = z(\eta_0)$ be an exact central point, with gap $\Delta_0 = \nu/\eta_0$. For $0 < \varepsilon < \Delta_0$, let $\mathcal{Z}_{\varepsilon}$ be the set of all strictly feasible primal–dual pairs with gap at most ε . Then*

$$\sqrt{\nu/2} \log \frac{\Delta_0}{\varepsilon} \leq d_{\hat{F}}(z_0, \mathcal{Z}_{\varepsilon}) \leq \sqrt{\nu} \log \frac{\Delta_0}{\varepsilon}. \tag{9.5}$$

The upper bound is the length of the central path from z_0 to $z(\nu/\varepsilon)$. Hence, for $0 < R < 1$, every forward- R sequence from z_0 to $\mathcal{Z}_{\varepsilon}$ has at least

$$\frac{\sqrt{\nu/2} \log(\Delta_0/\varepsilon)}{-\log(1-R)} \tag{9.6}$$

steps, even if intermediate points violate the affine equations.

Proof. At a center $z_c = z(\eta)$, conjugacy gives $F'_*(s_c) = -\eta x_c$ and $\widehat{F}(z_c) = \nu \log \eta - \nu$. For any feasible $z = (x, s)$, orthogonality of primal and dual feasible differences gives $\langle x - x_c, s - s_c \rangle = 0$. Hence

$$D\widehat{F}(z_c)[z - z_c] = -\eta\{\langle s_c, x - x_c \rangle + \langle x_c, s - s_c \rangle\} = -\eta\{\text{gap}(z) - \nu/\eta\}.$$

At $\eta = \nu/\varepsilon$, convexity proves that z_c minimizes $\widehat{F}$ on $\mathcal{Z}_\varepsilon$. In the product-cone metric, the dual norm of $D\widehat{F}$ is exactly $\sqrt{2\nu}$, since each logarithmically homogeneous barrier contributes ν to its square. Integrating this bound along any curve in the product cone proves the lower bound. Equation (9.1) gives the upper bound, and [lemma 2.1](#) proves (9.6). $\square$

This theorem is a quantitative restatement of [21, Theorem 5.1(c), Theorem 5.2, and Corollary 5.1], which already covers the whole set of feasible small-gap pairs and allows affinely infeasible intermediate points.

9.2 An optimal cone barrier with little primal movement

For blocks y_a of positive dimensions s_a , define the cone over a product of k Euclidean balls,

$$\mathcal{H} = \{(t, y_1, \dots, y_k) : t \geq \|y_a\|_2 \ (1 \leq a \leq k)\}.$$

Its classical barrier

$$\mathcal{F}(t, y) = -\sum_{a=1}^k \log(t^2 - \|y_a\|^2) + (k-1) \log t \quad (9.7)$$

has exact logarithmic parameter $\nu = k+1$. Because of the positive term $(k-1) \log t$, the usual combination rules do not prove self-concordance. Instead, place each y_a in row a of a k -row matrix Y , in its own column block. Then $YY^* = \text{diag}(\|y_a\|^2)$, and (9.7) is the restriction of the spectral-norm-cone barrier $(k-1) \log t - \log \det(t^2 I - YY^*)$ from [20, Proposition 5.4.6(i), p. 199]. Hence $\mathcal{F}$ is standard self-concordant with parameter $k+1$. Restricting further to one scalar direction in each block gives the $(k+1)$ -dimensional infinity-norm cone, whose barrier parameter is at least $k+1$ for $k \geq 2$ [13, Corollary 6.2]. For $k=1$, this section is the orthant $\mathbb{R}_+^2$, with optimal parameter two. Thus $k+1$ is also optimal among logarithmically homogeneous barriers on $\mathcal{H}$.

On the slice $t=1$, the barrier restricts to $\sum_a b(\|y_a\|)$, with exact parameter k by [proposition 3.2](#). For unit vectors u_a and weights $w_a \geq 0$, minimize $-\sum_a w_a \langle u_a, y_a \rangle$ on this slice. The central points are $y_a(\eta) = q(\eta w_a) u_a$, with the scalar function q of [section 4](#). Radial differentiation and (9.1) give

$$\nu_P(\eta) = q_{\text{eff}}(\eta) = \sum_a \left(1 - \frac{1}{\sqrt{1 + (\eta w_a)^2}}\right), \quad \nu_D(\eta) = k+1 - q_{\text{eff}}(\eta). \quad (9.8)$$

For $w_1 = 1$, all other weights zero, and $0 < \eta_0 < \eta_1$, the primal and dual central lengths L_P and L_D on $[\eta_0, \eta_1]$ satisfy

$$L_P \leq \log(\eta_1/\eta_0), \quad L_D \geq \sqrt{k} \log(\eta_1/\eta_0).$$

By [theorem 9.2](#), the full endpoint distance is at least $\sqrt{(k+1)/2} \log(\eta_1/\eta_0)$. Starting the primal path at its analytic center $\eta=0$ instead costs at most $1/\sqrt{2} + \log \eta_1$ for $\eta_1 \geq 1$, since $q_{\text{eff}}(\eta) \leq \eta^2/2$ on $(0, 1]$. Thus the barrier parameter alone gives no comparable primal lower bound.

This conclusion does not depend on the nonunique optimizer that zero weights create. Fix $0 < \varepsilon \leq 1$, write $\nu = k+1$, and take

$$w_1 = 1, \quad w_a = \varepsilon^2/\nu^2 \ (a \geq 2), \quad \eta_0 = 1, \quad \eta_1 = \nu/\varepsilon. \quad (9.9)$$

All weights are positive, so $(u_a)_a$ is the unique optimizer on the product of balls. The scalar inequality $1 - (1 + z^2)^{-1/2} \leq z^2/2$ gives, on this interval,

$$q_{\text{eff}}(\eta) \leq 1 + \frac{\varepsilon^2}{2\nu}, \quad L_P \leq \sqrt{1 + \frac{\varepsilon^2}{2\nu}} \log \frac{\nu}{\varepsilon}, \quad L_D \geq \sqrt{k - \frac{\varepsilon^2}{2\nu}} \log \frac{\nu}{\varepsilon}.$$

At η_1 , the full gap is exactly ε and the primal objective gap is at most ε . These objectives depend on ε , so the example rules out only a bound uniform over objectives. For a fixed objective with every $w_a > 0$, $q_{\text{eff}}(\eta) \rightarrow k$ as $\eta \rightarrow \infty$, so the small weights matter asymptotically for a fixed instance. [Proposition 5.1](#) and the decay examples of [section 5](#) describe this more precisely.

9.3 Three step counts on one sparse linear program

Theorem 9.3 (The cost of a feasible dual solution). *Take the dyadic data $w_i, a_i, T = a_r = \Theta(r)$ and $\varepsilon_r = re^{-T}$ of (6.1). Write the box linear program in the standard form*

$$u_i = 1 + x_i, \quad v_i = 1 - x_i, \quad u_i + v_i = 2, \quad \min \sum_i \left(-\frac{w_i}{2} u_i + \frac{w_i}{2} v_i \right). \quad (9.10)$$

The orthant barrier $\mathcal{F} = -\sum_i (\log u_i + \log v_i)$ has parameter $2r$ and restricts exactly to $U(x) = \sum_i b(x_i)$. Start at the primal-dual central point $z^c(0)$, where $s = \log \eta = 0$. For every fixed $0 < R < 1$, the least numbers of steps of forward- R sequences have the following orders as $r \rightarrow \infty$:

Metric and required endpoint	Number of steps
Restricted primal, arbitrary route from $x^c(0)$ to $x^c(T)$	$\Theta_R(r\sqrt{\log r})$
Restricted primal, exact central iterates between those endpoints	$\Theta_R(r \log r)$
Full primal-dual, any strictly feasible endpoint with gap $\leq 2\varepsilon_r$	$\Theta_R(r^{3/2})$
Full primal-dual, exact central route from $z^c(0)$ to $z^c(T)$	$\Theta_R(r^{3/2})$

The second order also holds under the weaker requirement of [theorem 6.2](#): iterates within a fixed distance of primal central points taken in any order, and an ε_r -accurate last iterate. The full primal-dual lower bound allows affinely infeasible intermediate points. The terminal point $x^c(T)$ is ε_r -accurate, and $z^c(T)$ has full gap exactly $2\varepsilon_r$.

Proof. Let α_i, β_i be the dual slacks associated with u_i, v_i . The signs in (9.10) give

$$\beta_i - \alpha_i = w_i, \quad u_i \alpha_i = v_i \beta_i = e^{-s}. \quad (9.11)$$

Consequently $x_i^c(s) = q(e^{s-a_i})$ and

$$\|\dot{z}_P^c\|^2 = q_{\text{eff}}(e^s), \quad \|\dot{z}_D^c\|^2 = 2r - q_{\text{eff}}(e^s) \geq r, \quad \|\dot{z}^c\|^2 = 2r. \quad (9.12)$$

The full central length on $[0, T]$ is exactly $\sqrt{2r} T$. The scalar error bound $w[1 - q(\eta w)] \leq 1/\eta$ proves terminal primal accuracy; summing (9.11) gives full gap $2re^{-T} = 2\varepsilon_r$.

Next we bound the distance to every feasible small-gap pair directly. At the starting center, $1/\alpha_i^0 + 1/(\alpha_i^0 + w_i) = 2$; since $w_i \leq 1$ and the left side is decreasing, $\alpha_i^0 \geq 1/\sqrt{2}$. At every feasible pair,

$$\text{gap} = \sum_i (u_i \alpha_i + v_i \beta_i) = \sum_i (2\alpha_i + v_i w_i).$$

Thus $\text{gap} \leq 2\varepsilon_r$ forces every $\alpha_i \leq \varepsilon_r$. The dual orthant metric is Euclidean in coordinatewise logarithms. Projecting an arbitrary curve in the full space onto the logarithms of these r slacks proves

$$d_{\widehat{\mathcal{F}}}(z^c(0), \mathcal{Z}_{2\varepsilon_r}) \geq \sqrt{r} [T - \log r - \tfrac{1}{2} \log 2]_+. \quad (9.13)$$

With the central upper bound and [lemma 2.1](#), this proves both primal–dual rows, with no prescribed final dual point. [Theorem 9.2](#) gives another lower bound of this order.

For the primal rows, [theorem 6.1](#) measures from the analytic center to the first accurate parameter $s_\varepsilon \in (T - 1, T]$. The negative-parameter part of the central path has length $O(1)$: the scalar bound $v(s - a_i) \leq e^{s - a_i} / \sqrt{2}$ bounds its length by $(\sum_i w_i^2 / 2)^{1/2} = O(1)$. The last segment from s_ε to T has length at most $\sqrt{r}$. By the triangle inequality, the primal endpoint distance and central length between $s = 0$ and $s = T$ remain $\Theta(r\sqrt{\log r})$ and $\Theta(r \log r)$. The endpoint distance and a cover of the central path by bounded chords prove the arbitrary-route row and the central upper bound. For the central lower bound, prepend a bounded number of such chords on the negative-parameter part and apply the lower bound for step sequences near the central path, [theorem 6.2](#). The same argument covers iterates near the path with central points in any order. It uses the potential P from that proof, not central arclength alone. $\square$

By the count after [\(6.1\)](#), the data have rational encoding length $B = \Theta(r^2)$, so the three step counts are

$$\Theta(\sqrt{B \log B}), \quad \Theta(\sqrt{B} \log B), \quad \Theta(B^{3/4}). \quad (9.14)$$

Requiring a feasible dual solution multiplies the least step count in this metric by $\Theta(\sqrt{r / \log r})$ compared with the best primal route, although each equality row has two nonzeros and each column one. The comparison counts bounded local steps; neither this theorem nor the speed identity is a lower bound on running time or oracle queries.

10 What a change of formulation can preserve

[Theorem 9.3](#) concerns one formulation and one barrier. This section gives two movement lower bounds for other formulations, including those with auxiliary variables. For log-determinant barriers of symmetric cones, the rank of an exposing dual solution bounds primal movement; for every logarithmically homogeneous barrier, a bound on the factor dimensions bounds full primal–dual movement. Neither statement derives a primal lower bound from an upper bound on the barrier parameter.

10.1 Exposing dual solutions and free auxiliary variables

Let $K = \prod_i K_i$ be a product of symmetric cones in Euclidean Jordan algebras with trace inner products, and let $\Omega = \{x \in \text{int } K : Ax = b\} \neq \emptyset$. Suppose the objective ℓ has finite supremum ℓ_* over Ω and an attained dual solution $s = (s_i) \in K$ that exposes the optimum:

$$\ell_* - \ell(x) = \sum_i \langle s_i, x_i \rangle \quad (x \in \Omega). \quad (10.1)$$

Let c_i be the support idempotent of s_i , let $q_i = \text{rank}(s_i)$, and omit factors of rank zero. Here $P(c_i)$ is the orthogonal Peirce projection onto the algebra with unit c_i , and $\det_{c_i}$ is its determinant.

Theorem 10.1 (Distance bound from the rank of an exposing dual solution). *Use the barrier*

$$F(x) = - \sum_i \alpha_i \log \det_{c_i} x_i, \quad \alpha_i \geq 1,$$

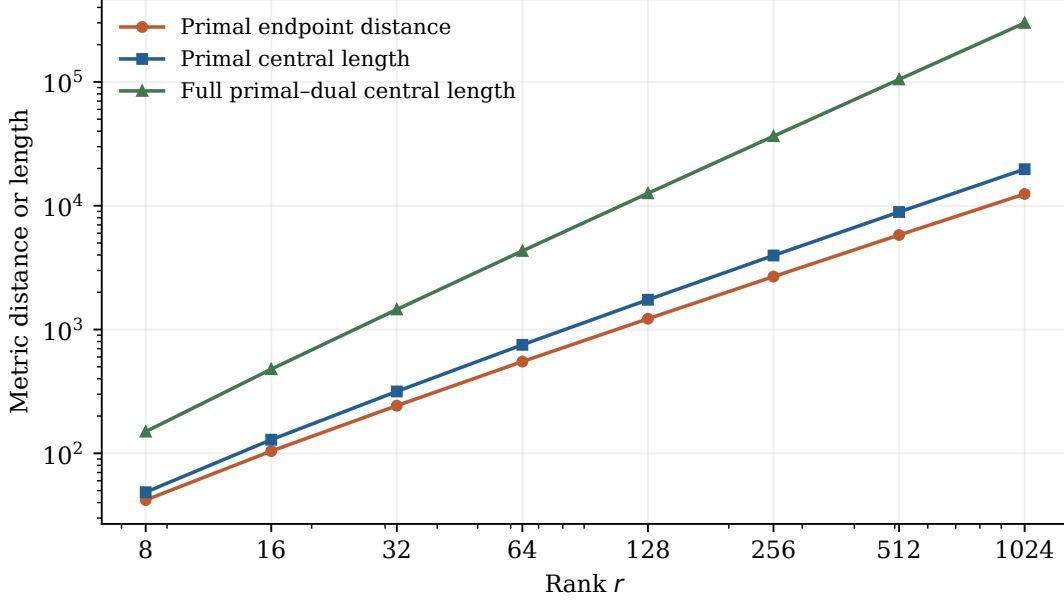


Figure 2: Numerical values for the dyadic family (6.1), from $s = 0$ to $T = a_r$. The curves show the primal endpoint distance $d_U(x^c(0), x^c(T))$ (exact formula), the primal central arclength on $[0, T]$ (quadrature), and the full central length $\sqrt{2r}T$. The first curve is not the distance to the whole accurate set, and the third is not the distance to the set of feasible small-gap pairs. The orders $r\sqrt{\log r}$, $r \log r$, and $r^{3/2}$, and the corresponding step counts, follow from theorems 6.1 and 9.3. Data and generation code accompany the paper.

restricted to Ω . Put $Q_\alpha = \sum_i \alpha_i q_i > 0$ and, for any reference $x^c \in \Omega$, define

$$\Delta_{c,\alpha} = Q_\alpha \left[\prod_i \frac{\{\det_{c_i}(s_i) \det_{c_i}(P(c_i)x_i^c)\}^{\alpha_i}}{\alpha_i^{\alpha_i q_i}} \right]^{1/Q_\alpha}. \quad (10.2)$$

For $\varepsilon > 0$, every feasible y with objective gap at most ε satisfies

$$d_F(x^c, y) \geq \sqrt{Q_\alpha} [\log(\Delta_{c,\alpha}/\varepsilon)]_+. \quad (10.3)$$

The bound does not restrict eigenvalues outside the support of s or auxiliary variables, and needs no strict complementarity. If a method starts at x_0 with $d_F(x_0, x^c) \leq D_0$ and reaches such a y after N rounds, each moving the iterate a distance at most $B_0 > 0$, then

$$N \geq \frac{[\sqrt{Q_\alpha} \log(\Delta_{c,\alpha}/\varepsilon) - D_0]_+}{B_0}. \quad (10.4)$$

In particular, if $\|x_0 - x^c\|_{x^c} \leq \rho$ with $0 \leq \rho < 1$ and each round has at most $m \in \mathbb{N}$ forward R -chords, $0 < R < 1$, one can take $D_0 = -\log(1 - \rho)$ and $B_0 = -m \log(1 - R)$.

Proof. We give the compression argument in full to fix the metric normalization, including the exceptional Jordan algebra. For one idempotent c of rank q , put $a = P(c)x$ and $\phi_c(x) = -\log \det_c a$. The Hessian of $-\log \det x$ is $P(x^{-1})$ with inverse $P(x)$, where $P(u) = 2L_u^2 - L_{u^2}$ is the quadratic representation and $L_u v = u \circ v$. Since a^{-1} belongs to the Peirce algebra,

$$d\phi_c(x)[h] = -\langle a^{-1}, h \rangle, \quad \|d\phi_c(x)\|_{x,*}^2 = \langle a^{-1}, P(x)a^{-1} \rangle = q.$$

Indeed, the fundamental identity $P(P(u)v) = P(u)P(v)P(u)$ with $u = c$ gives $P(c)P(x)P(c) = P(a)$; then $P(a)a^{-1} = a$ and $\langle a^{-1}, a \rangle = q$. The compression a is positive definite in its Peirce algebra: if $x \succeq \delta e$, positivity of $P(c)$ gives $a \succeq \delta c$. For $\Phi = \sum_i \alpha_i \phi_{c_i}$, the product metric therefore gives $\|d\Phi\|_{F,*}^2 = Q_\alpha$ before affine restriction and at most Q_α afterwards. These quadratic-representation identities are standard [10]; we do not claim the principal-minor metric identity as new.

Set $g_i = \langle s_i, y_i \rangle > 0$ on the active factors. Inside the Peirce algebra, $z_i = P(s_i^{1/2})P(c_i)y_i$ has trace g_i and determinant $\det_{c_i}(s_i) \det_{c_i}(P(c_i)y_i)$. Spectral AM–GM gives

$$\det_{c_i}(P(c_i)y_i) \leq \det_{c_i}(s_i)^{-1} (g_i/q_i)^{q_i}.$$

Under $\sum_i g_i \leq \varepsilon$, the weighted product $\prod_i (g_i/q_i)^{\alpha_i q_i}$ is maximized at $g_i = \varepsilon \alpha_i q_i / Q_\alpha$. Thus

$$\prod_i \det_{c_i}(P(c_i)y_i)^{\alpha_i} \leq (\varepsilon/Q_\alpha)^{Q_\alpha} \prod_i \alpha_i^{\alpha_i q_i} \det_{c_i}(s_i)^{-\alpha_i}.$$

Comparing with x^c proves $\Phi(y) - \Phi(x^c) \geq Q_\alpha \log(\Delta_{c,\alpha}/\varepsilon)$. Integrating this dual-norm bound along any feasible curve proves (10.3). The triangle inequality and lemma 2.1 prove the round statements. $\square$

For unit weights this gives $Q = \sum_i q_i$ and $\Delta_c = Q[\prod_i \det_{c_i}(s_i) \det_{c_i}(P(c_i)x_i^c)]^{1/Q}$. The coefficient $\sqrt{Q_\alpha}$ is asymptotically sharp on the full cone. Apply a quadratic-representation isometry sending the reference point to the unit element, diagonalize the transformed dual solution, and multiply its support coordinates by t , keeping the others equal to one. The gap is $t\langle s, e \rangle$ and the curve has length $\sqrt{Q_\alpha} \log(1/t)$. Together with (10.3), this proves the exact leading coefficient as $\varepsilon \downarrow 0$. An affine slice need not contain this path; we claim no sharp upper bound on an arbitrary slice.

Hauser and Güler’s classification [12, Theorem 5.5] states that every self-scaled barrier on a symmetric cone is, up to an additive constant, a sum of these log-determinants on its irreducible factors with coefficients $\alpha_i \geq 1$. Thus theorem 10.1 covers every such barrier, with the weighted scale $\Delta_{c,\alpha}$; replacing Q by Q_α but keeping the unweighted Δ_c would be incorrect. The bound depends on the rank of s ; a general lower bound on the ranks attainable by different lifted formulations is a separate question, which the theorem neither assumes nor answers.

10.2 A bound from factor dimensions for arbitrary cone barriers

Theorem 10.2 (Bounded factor dimension). *Let a compact full-dimensional convex body $C \subset \mathbb{R}^D$, $D \geq 1$, have an exact affine lift $C = \pi(K \cap \mathcal{L})$, where K is a finite product of proper cones, $\mathcal{L}$ is an affine subspace, and π is an affine map. Let the minimal face be the smallest face of K containing $K \cap \mathcal{L}$; it is a product of faces of the factors of K . Regarding this product as a cone in its linear span and removing its zero factors gives the reduced product cone. Suppose each of its factors has dimension at most an integer $d \geq 2$. For $h \geq 1$ define*

$$\Psi_d(h) = 2\lfloor h/d \rfloor + \min\{h \bmod d, 2\}.$$

Every logarithmically homogeneous standard self-concordant barrier on the reduced product cone, including one that couples the factors, satisfies

$$\nu \geq \Psi_d(D+1) \geq \frac{2(D+1)}{d}. \quad (10.5)$$

For a strictly feasible primal–dual conic pair on this cone, consider routes from an exact central point with gap Δ_0 to a strictly feasible pair with gap at most ε , $0 < \varepsilon < \Delta_0$. If each round has at most $m \in \mathbb{N}$ forward R -chords in the product metric, $0 < R < 1$, the number of rounds is at least

$$\frac{\sqrt{\Psi_d(D+1)/2} \log(\Delta_0/\varepsilon)}{-m \log(1-R)}. \quad (10.6)$$

Proof. Let M be the dimension of the span of the reduced product cone. The affine projection of its feasible slice spans $\mathbb{R}^D$, so $M \geq D$. If $M = D$, the linear part of π is invertible on this span, so the feasible slice has full affine hull and the affine equations impose no restriction: the feasible slice is the whole reduced cone. Its invertible affine image is unbounded, contrary to compactness. Therefore $M \geq D + 1$. This elementary argument is consistent with the classical slack-factorization framework [11], but needs no choice of boundary certificates.

Let r_0 and q_0 be the numbers of ray factors and of factors of dimension at least two. In each nonray factor choose a two-dimensional linear plane containing an interior point. Its cone section is proper and full-dimensional in the plane, hence linearly isomorphic to $\mathbb{R}_+^2$; its relative interior lies in the factor's interior and its boundary in the factor's boundary. Together with all rays, these sections give an orthant section of dimension $r_0 + 2q_0$ through the interior of the reduced cone. Restriction preserves standard self-concordance, the degree of logarithmic homogeneity, and boundary blow-up. The classical orthant parameter lower bound [13, 20] therefore gives $\nu \geq r_0 + 2q_0$, even when the original barrier couples factors. The dimension bound d gives $r_0 + dq_0 \geq M \geq D + 1$.

To solve the integer minimization, write $h = ad + b$, $0 \leq b < d$. For $q_0 \leq a$ the smallest possible value of $r_0 + 2q_0$ is $2q_0 + h - dq_0 = h - (d - 2)q_0$, minimized at $q_0 = a$ and equal to $2a + b$. For $q_0 \geq a + 1$ this value is at least $2a + 2$, attained with no rays. The minimum is therefore $2a + \min(b, 2)$; the case $b = 0$ is attained with $q_0 = a$. Since $\min(b, 2) \geq 2b/d$, this proves (10.5). Apply theorem 9.2 for (10.6). $\square$

The parameter bound concerns barriers on the reduced product cone, not arbitrary barriers on C or on the affine slice. The theorem combines classical restriction, dimension, and primal-dual arguments; it does not determine the least dimension of a lift. For a product of balls, $D = \sum_a s_a$, so the bound uses the total ball dimension and no curvature or regularity assumptions.

10.3 Grouped balls, norm trees, and PSD packing

The following formulations show that fewer conic blocks need not reduce movement in the given barrier metric. Full proofs, including all auxiliary variables, are in appendix C.

Take k unit balls, a unit vector c_a for each, and the objective $\ell = \sum_a \langle c_a, w_a \rangle$ with maximum k . For a grouped paraboloid lift, partition each ball's coordinates into h groups, introduce $q_{aG} = s_{aG} - \|w_{aG}\|^2 > 0$ with $\sum_G s_{aG} = 1$, and use $F = -\sum_{a,G} \log q_{aG}$. For a PSD packing, place the $H = kh$ group vectors as columns into real matrix blocks W_ℓ in any arrangement, let every off-diagonal entry of S_ℓ be free, and use

$$D_\ell = S_\ell - W_\ell^T W_\ell \succ 0, \quad F = -\sum_\ell \log \det D_\ell, \quad \sum_{\gamma \text{ from ball } a} (S_{\ell(\gamma)})_{\gamma\gamma} = 1.$$

A group column may be padded with zeros; its free entries are its group's coordinates. Each group occurs exactly once. Both barriers are standard self-concordant with exact restricted parameter H ; for each, the central path $z(\tau) = \arg \min(F - \tau\ell)$ has $w_a = r(\tau)c_a$, with

$$q(\tau) = \frac{2}{\sqrt{h^2 + \tau^2} + h}, \quad r(\tau) = \frac{\tau}{\sqrt{h^2 + \tau^2} + h}, \quad \left\| \frac{dz}{d \log \tau} \right\|^2 = H \left(1 - \frac{h}{\sqrt{h^2 + \tau^2}} \right). \quad (10.7)$$

On the path, $q_{aG} = q(\tau)$ (grouped lift) and $D_\ell = q(\tau)I$ (packing). For every $\tau_0 \geq 0$ and every feasible y with gap at most ε ,

$$d_F(z(\tau_0), y) \geq \sqrt{H} [\log(q(\tau_0)H/(2\varepsilon))]_+. \quad (10.8)$$

At the analytic center $\tau_0 = 0$ the argument of the logarithm is $k/(2\varepsilon)$, independent of the packing and its off-diagonal variables.

For a rooted norm tree with b internal nodes, each with at least two children, use one Lorentz-cone block per internal node:

$$q_{av} = t_{av}^2 - \sum_{u \in I(v)} t_{au}^2 - \sum_{j \in J(v)} w_{aj}^2 > 0, \quad t_{av} > 0, \quad t_{a,\text{root}} = 1. \quad (10.9)$$

Here $I(v)$ and $J(v)$ are the internal and leaf children of v . The positive sheet $t_{av} > 0$ is needed for convexity. Use $F = -\sum_{a,v} \log q_{av}$ and put $L = kb$. Its exact parameter is $\nu = k\nu_T$, where

$$\nu_T = \begin{cases} 2b - 2, & \text{the root has only internal children,} \\ 2b - 1, & \text{the root has a leaf child.} \end{cases} \quad (10.10)$$

Its central path is (10.7) with $h = b, H = L$, regardless of the tree shape. If n_v counts internal nodes in the subtree at v , including v , and $C_{av} = \sum_{j \text{ below } v} c_{aj}^2$, the internal coordinates are $t_{av}^2 = n_v q + r^2 C_{av}$. At the analytic center $\bar{z}$, $q_{av} = 1/b$, and every feasible y with gap at most ε satisfies the sharper barrier-value bound

$$d_F(\bar{z}, y) \geq \frac{L}{\sqrt{\nu}} [\log(k/(2\varepsilon))]_+ \geq \sqrt{L/2} [\log(k/(2\varepsilon))]_+. \quad (10.11)$$

Appendix C proves the exact parameter (10.10) by a metric calculation; simply subtracting the fixed root's contribution from the cone parameter would not be justified.

For all three formulations, the central path from the analytic center to objective gap ε has length

$$\sqrt{M_0} \rho(1 - \varepsilon/k) = \sqrt{M_0} \log(k/\varepsilon) + O(\sqrt{M_0}), \quad M_0 = H \text{ or } L, \quad (10.12)$$

for $0 < \varepsilon < k$, with the asymptotic form as $\varepsilon/k \downarrow 0$. Sampling this path and the distance bounds give matching forward R -chord counts $\Theta_R(\sqrt{M_0} \log(k/\varepsilon))$, uniformly for $0 < \varepsilon \leq k/4$. An approximate starting point and a fixed number of chords per round are handled as in (10.4). These statements concern the given barrier metrics; the block count alone implies none of these primal lower bounds.

11 Interpretation

Our results separate three step counts: the steps to an accurate point by any route, the extra steps needed to stay near the central path, and the extra steps needed to reach a feasible dual solution. The objective weights determine when the coordinates, singular values, or eigenvalues start to move. Because they start in turn, the central path can be longer than a route that moves them together toward an accurate point. Requiring a feasible dual solution can add movement absent from the primal path. On the dyadic linear program, these effects have different asymptotic orders with matching bounds, in length and in bounded local steps.

The exact distance to the accurate set matters because the endpoint of the central path need not be the closest accurate point. It also shows why a single rank or determinant bound may not suffice: the whole distribution of objective weights enters the exact distance. Conversely, the sharp bound of theorem 4.1 limits how much longer a central arc can be than the distance between its endpoints. With equal scales, the worst ratio is of order $\sqrt{\log r}$ in the objective rank r , even when the matrix space is much larger.

A new barrier or formulation requires a new comparison in its own metric. An optimal barrier parameter alone does not remove the extra central-path length, as the universal and entropic cube

barriers and the coupled radial barriers show. A cone barrier restricted to a slice can also have a much smaller primal speed than the full parameter suggests. [Theorem 10.1](#) and the concrete formulations give direct lower bounds with freely moving auxiliary variables, instead of inferring primal cost from a parameter bound.

These conclusions assume a fixed barrier metric, a specified starting point, a required final accuracy or feasible duality gap, and steps bounded in the local norm at their start. Under them, the explicit shortest routes and matching lower bounds separate the cost of staying near the central path from the cost of a feasible dual solution.

A Rational certificates for $c_\star$

This appendix proves the strict bounds in [\(4.9\)](#). The standalone script `scripts/verify_scalar_certificate.py`, which uses only the Python standard library, repeats all finite arithmetic checks below; the proof uses no floating-point evaluation.

A.1 An elasticity bound

For $x = \rho^{-1}(y) > 0$, define

$$E(y) = \frac{yp'(y)}{p(y)}, \quad K(y) = \frac{y^2(1-x^2)}{2x^2(1+x^2)}.$$

Direct differentiation using $p = b' \circ \rho^{-1}$ gives

$$\frac{dE}{d \log y} = E - K, \quad E \geq 1, \quad 0 < K < 1. \quad (\text{A.1})$$

For the last inequality, set $B(x) = \sqrt{2x}\sqrt{(1+x^2)/(1-x^2)}$, so $K = (\rho(x)/B(x))^2$. With $u = x^2$, the derivative ratio is

$$\frac{B'}{\rho'} = \frac{1+2u-u^2}{(1+u)\sqrt{1-u}} > 1,$$

because the difference of squared numerator and denominator is $u(3+3u-3u^2+u^3) > 0$. Equality of the two functions at zero proves $\rho < B$. Convexity of p gives $E \geq 1$, so E increases strictly. Solving the differential inequality in [\(A.1\)](#) on a multiplicative ray gives $E(ty) > 1 + t(E(y) - 1)$ for $t > 1$. Consequently, for every $a > 1$,

$$\log \frac{p(ay)}{yp'(y)} > h_1(E) := \log a + (a-1)(E-1) - \log E. \quad (\text{A.2})$$

For a stronger bound when $E > 2$, concavity of the square root gives

$$\rho(x) \leq \sqrt{2}D(x), \quad D(x) = \frac{3}{2} \operatorname{artanh} x - x/2.$$

At $x = 4/5$, $D = (3/2) \log 3 - 2/5 < 2497/2000$. Indeed $\log 3 < 1099/1000$ follows by summing the exponential Taylor polynomial through degree six. Hence $E(\rho(4/5)) < 2497\sqrt{41}/8000 < 2$, the latter inequality being $2497^2 \cdot 41 < 64 \cdot 2000^2$. Thus $E(y) > 2$ implies $x > 4/5$. On this region,

$$K \leq G(x) := \frac{D(x)^2(1-x^2)}{x^2(1+x^2)}.$$

The positive power series for artanh gives $D(x) \geq x(1 + x^2/2)$, and $D'(x) = (1 + x^2/2)/(1 - x^2)$. It follows that

$$\frac{1}{2}(\log G)' = \frac{D'}{D} - \frac{x}{1 - x^2} - \frac{1}{x} - \frac{x}{1 + x^2} \leq -\frac{x}{1 + x^2} < 0.$$

Therefore

$$K < G(4/5) < \left(\frac{2497}{2000}\right)^2 \frac{225}{656} = \frac{56115081}{104960000} < k := \frac{107}{200}.$$

Since E increases, this bound holds on the whole ray ty , $t > 1$, once $E(y) > 2$. Integrating (A.1) now gives $E(ty) > k + t(E(y) - k)$ for $t > 1$, and hence

$$\log \frac{p(ay)}{yp'(y)} > h_k(E) := k \log a + (a - 1)(E - k) - \log E \quad (E > 2). \quad (\text{A.3})$$

A.2 The strict upper certificate

Take $a = 69/50$. On $1 \leq E \leq 2$, h_1 is decreasing, and

$$h_1(E) \geq h_1(2) = \log(69/100) + 19/50 > 0,$$

because $e^{19/50} > 1 + 19/50 + (19/50)^2/2 = 7261/5000 > 100/69$. On $E > 2$, the convex function h_k has its global minimum at $E = 50/19$, where its value is

$$\log(19/50) + 1 + \frac{107}{200}(\log(69/50) - 19/50).$$

The positive series $\log((1 + z)/(1 - z)) = 2 \sum_{j \geq 0} z^{2j+1}/(2j + 1)$ yields, with $r_0 = 31/69$ and $s_0 = 19/119$,

$$\begin{aligned} \log(19/50) &> -2 \left(r_0 + \frac{r_0^3}{3} + \frac{r_0^5}{5(1 - r_0^2)} \right) = -\frac{9064603453}{9362506500}, \\ \log(69/50) &> 2(s_0 + s_0^3/3) = \frac{1628072}{5055477}. \end{aligned}$$

Substitution leaves the strictly positive rational lower bound

$$h_k(50/19) > \frac{255839420916449}{315546241820670000} > 0. \quad (\text{A.4})$$

Thus $p((69/50)y) > yp'(y)$ for every $y > 0$. Since the maximum defining $c_\star$ is attained, the upper bound on $c_\star$ is strict.

A.3 The strict lower certificate

The elementary transform in (4.11) has series

$$Y(z) = \sum_{j \geq 0} \frac{(2 - 2^{-j})z^{2j+1}}{2j + 1}, \quad S_N(z) < Y(z) < S_N(z) + \frac{2z^{2N+3}}{(2N + 3)(1 - z^2)}. \quad (\text{A.5})$$

Here S_N includes $j = 0, \dots, N$; the tail estimate follows by bounding $2 - 2^{-j} < 2$ and every tail denominator by $2N + 3$. Set

$$v_0 = \frac{3787}{4000}, \quad w_0 = \frac{9797031}{10^7}, \quad a_0 = \frac{68743}{50000}.$$

Summing through $N = 1200$ in exact rational arithmetic, with rational squaring for A , gives

$$\begin{aligned} \frac{2453756741730599}{10^{15}} &< Y(v_0) < \frac{12268783708653}{5 \cdot 10^{12}}, & \frac{134943211257327}{4 \cdot 10^{13}} &< Y(w_0), \\ A(v_0) &> \frac{405367307458211}{4 \cdot 10^{13}}, & A(w_0) &< \frac{25381942601625079}{10^{15}}. \end{aligned}$$

These finite enclosures imply

$$\begin{aligned} Y(w_0) - a_0 Y(v_0) &> \frac{2071874360571}{250000000000000000} > 0, \\ Y(v_0)A(v_0) - w_0 A(w_0) &> \frac{2049519399569150216498389}{400000000000000000000000000000000} > 0. \end{aligned} \quad (\text{A.6})$$

The second inequality states $P(w_0) < Y(v_0)A(v_0)$. Since P and Y are strictly increasing, it follows that $Y(W(v_0)) > Y(w_0) > a_0 Y(v_0)$. Hence v_0 is a strictly feasible witness for $c_\star > a_0$. The proof uses only the interval bounds, the tail estimate, and rational squaring; it does not rely on a numerical optimizer or on uniqueness of a stationary point.

A.4 Location of every maximizer

The same inequalities also locate every maximizer in the definition (4.9) of the constant $c_\star$ for the scalar barrier b . Keep $a_0 = 68743/50000$ and $k = 107/200$ in equations (A.2) and (A.3). Exact positive-series bounds for logarithms give

$$\begin{aligned} h_1(2) &> 2589087717927/40000000000000000 > 0, \\ h_k(571/250) &> 2011885403423/10000000000000000 > 0, \\ h_k(773/250) &> 8579961311243/50000000000000000 > 0. \end{aligned}$$

These bounds are checked in finite rational arithmetic: for rational $q \in (0, 1)$, the partial sum of $2 \sum_{j \geq 0} q^{2j+1}/(2j+1)$ through $j = N$ bounds $\log((1+q)/(1-q))$ below, and adding $2q^{2N+3}/[(2N+3)(1-q^2)]$ bounds it above. Taking $q = (z-1)/(z+1)$ for $z = a_0, 2, 571/250, 773/250$ and $N = 200$ verifies the displayed inequalities using rational arithmetic. The function h_1 decreases on $[1, 2]$. The convex function h_k decreases until $E = 1/(a_0 - 1)$ and increases thereafter; this minimum lies strictly between $571/250$ and $773/250$. Therefore the ratio defining $c_\star$ can exceed a_0 only for $571/250 < E < 773/250$.

In the coordinate of (4.11), $E = Y(v)/v$ is strictly increasing by (A.1). The series (A.5) through $N = 1200$ gives

$$\begin{aligned} \frac{571}{250} \frac{4611}{5000} - Y(4611/5000) &> 159652343530451/200000000000000000 > 0, \\ Y(9701/10000) - \frac{773}{250} \frac{9701}{10000} &> 19492543073837/250000000000000000 > 0. \end{aligned}$$

Since every global maximizer has ratio $c_\star > a_0$, these inequalities, followed by rational squaring in $x = v/\sqrt{2-v^2}$, prove

$$\frac{4611}{5000} < v_\star < \frac{9701}{10000}, \quad \frac{43}{50} < x_\star < \frac{943}{1000}. \quad (\text{A.7})$$

Finally, differentiating $P(W(v)) = Y(v)A(v)$ shows that every stationary point satisfies

$$Y(v)[Y'(v)A(v) + Y(v)A'(v)] = Y(W(v))Y'(v)A(W(v)), \quad A'(v) = \frac{v(3-v^2)}{(1-v^2)^2\sqrt{2-v^2}}.$$

These location bounds do not imply that the stationary point, or the maximizer in the definition of $c_\star$, is unique.

B The scalar envelope and a barrier with nonmonotone speed

Proposition B.1 (Exact envelope for a relaxed problem with constant $k = \log 2$). *Let $p \in C^2([0, \infty))$ satisfy $p(0) = 0$, $p' > 0$, $|p''| \leq p'$, and $p \leq p'$ on $[0, \infty)$. Put $k = \log 2$ and*

$$C(a) = \begin{cases} 1 + \frac{1}{a} \log \frac{k}{k(2 - e^{-a}) - a}, & 2a \leq k(2 - e^{-a}), \\ 1 + \frac{1}{a} \log \frac{4a}{k(2 - e^{-a})^2}, & 2a \geq k(2 - e^{-a}). \end{cases} \quad (\text{B.1})$$

Then $C_{\text{sc}} = \max_{a>0} C(a)$ has a unique maximizer, is strictly below two, and

$$a \leq p^{-1}(ap'(a)/k) \leq C(a)a \leq C_{\text{sc}}a.$$

For the C^2 function p , put $v = p/p'$. The function $C(a)$ is the exact envelope when v is relaxed to any locally absolutely continuous function $v : [0, \infty) \rightarrow [0, 1]$ satisfying

$$v(0) = 0, \quad 0 < v(y) \leq 1 \quad (y > 0), \quad 1 - v \leq v' \leq 1 + v \quad a.e. \quad (\text{B.2})$$

This exactness holds for the fixed constant k ; it is not a claim about the optimal arclength constant for products of copies of one smooth barrier.

Proof. The derivative identity $v' = 1 - vp''/p'$ gives (B.2). Integration yields

$$1 - e^{-a} \leq v(a) \leq \min(1, e^a - 1), \quad \frac{v(a)}{a} \leq \frac{1}{k}.$$

For the last inequality, $(e^a - 1)/a$ increases and $1/a$ decreases; their crossing is at $a = k$. Thus the comparison point $a + t$, defined by $p(a + t) = ap'(a)/k$, has $t \geq 0$. Write $u = v(a)$ and integrate $(\log p)' = 1/v$:

$$\int_0^t \frac{dq}{v(a + q)} = \log \frac{a}{ku}. \quad (\text{B.3})$$

The upper bound $v' \leq 1 + v$ and the cap $v \leq 1$ give $v(a + q) \leq m_u(q) := \min\{1, (1 + u)e^q - 1\}$. The cap is reached at $q_u = \log(2/(1 + u))$. For $t \leq q_u$,

$$\int_0^t \frac{dq}{m_u(q)} = \log \frac{(1 + u) - e^{-t}}{u}.$$

For $t \geq q_u$, the integral equals $\log((1 + u)/(2u)) + t - q_u$. The right side of (B.3) is fixed, and the actual integral is at least this envelope integral. Solving the latter therefore gives an upper bound on t :

$$t \leq \begin{cases} \log \frac{k}{k(1 + u) - a}, & 2a \leq k(1 + u), \\ \log \frac{4a}{k(1 + u)^2}, & 2a \geq k(1 + u). \end{cases}$$

Both expressions decrease with u , and their values and first derivatives agree at the switch. Substitute $u \geq 1 - e^{-a}$ to obtain (B.1).

The two branches meet at exactly one positive point: $2a - k(2 - e^{-a})$ has positive derivative $2 - ke^{-a}$ and changes sign. It lies below k . On the first branch, $C(a) < 2$ is equivalent to $2k(1 - e^{-a}) - a > 0$. This strictly concave function vanishes at 0 and k , and the branch has $0 < a < k$. On the second branch, $C(a) < 2$ is equivalent to

$$k(4e^a - 4 + e^{-a}) - 4a > 0.$$

Since $k > 2/3$, the expression exceeds $(2/3)(4e^a + e^{-a} - 4 - 6a)$. The inequalities $e^a \geq 1 + a + a^2/2 + a^3/6$ and $e^{-a} \geq 1 - a$ bound the parenthesis below by $1 - 3a + 2a^2 + (2/3)a^3$. Its minimum on $[0, \infty)$ is $(16 - 5\sqrt{10})/3 > 0$, as differentiation shows. Here $k > 2/3$ follows, for example, from the first term of the positive series for $\log 2 = 2 \operatorname{artanh}(1/3)$.

The two branches are continuously differentiable at their common boundary. Their endpoint limits are $1/k$ as $a \downarrow 0$ and 1 as $a \rightarrow \infty$. Moreover $C(k) = 1 + k^{-1} \log(16/9) > 1/k$, because $\log(32/9) > 1$ (use $e < 3 < 32/9$). Hence continuity proves a maximum at a finite positive argument, and the strict pointwise bound implies $C_{\text{sc}} < 2$. To prove uniqueness and to evaluate C_{sc} numerically, write $C = 1 + B/a$, where B is the logarithm on the appropriate branch. An interior stationary point satisfies $aB' = B$, with

$$B' = \frac{1 - ke^{-a}}{k(2 - e^{-a}) - a} \quad \text{or} \quad B' = \frac{1}{a} - \frac{2e^{-a}}{2 - e^{-a}}.$$

On the first branch, $B = -\log(2 - e^{-a} - a/k)$ is strictly convex, since its positive logarithm argument has second derivative $-e^{-a} < 0$. As $B(0) = 0$, $aB' - B > 0$ there. On the second,

$$B'' = -\frac{1}{a^2} + \frac{4e^{-a}}{(2 - e^{-a})^2} < 0.$$

Indeed the strict inequality is equivalent to $2e^{a/2} - e^{-a/2} > 2a$; the Taylor bounds $e^t \geq 1 + t + t^2/2$ and $e^{-t} \leq 1 - t + t^2/2$ for $t \geq 0$ give a difference at least $1 - a/2 + a^2/8 > 0$. Thus $aB' - B$ strictly decreases on the second branch, starts positive by continuous differentiability at the switch, and tends to $-\infty$. It has one zero, which is the unique global maximizer. Numerical evaluation gives $a \approx 0.65011438345$ and $C_{\text{sc}} \approx 1.831856423$; these decimals are not used in the bound or in its attainment proof.

Finally, for fixed a take $v(y) = 1 - e^{-y}$ until a , then follow $v' = 1 + v$ until reaching one, and remain one thereafter. This continuous, piecewise differentiable function obeys (B.2) almost everywhere and attains every envelope equality used at a . Reconstruct p by $(\log p)' = 1/v$, with normalization $p'(0) = 1$; the behavior $v(y) = y + O(y^2)$ gives a well-defined positive solution with $p(0) = 0$. This proves exactness for the stated relaxed problem. Likewise, k is the largest constant valid for all admissible functions v , since the admissible function $v = \min(1, e^y - 1)$ attains $\inf_y y/v(y) = k$. These two extremal functions need not come from one fixed smooth scalar barrier. $\square$

B.1 A smooth barrier whose central speed is not monotone

For $A > 4$ define, for all real y ,

$$P_A(y) = \frac{1 + 2e^{-A} \cosh y}{1 + 2e^{-A}}, \quad X_A(y) = \int_0^y \frac{du}{P_A(u)}, \quad Z_A(y) = \int_0^y P_A(u) du.$$

The map X_A is a smooth increasing bijection from $\mathbb{R}$ onto a finite interval $(-b_A, b_A)$: $P_A(y) \geq e^{y-A}/(1 + 2e^{-A})$ for $y \geq 0$, and P_A is even. Define f_A by $f'_A(X_A(y)) = Z_A(y)$ and $f_A(0) = 0$. All these functions are smooth, so no piecewise smoothing or boundary extension is needed. Differentiation gives

$$f''_A(X_A(y)) = P_A(y)^2, \quad \frac{|f'''_A|}{(f''_A)^{3/2}} = 2 \frac{|P'_A|}{P_A} < 2, \quad V_A(y) := \frac{f'_A}{\sqrt{f''_A}} = \frac{y + 2e^{-A} \sinh y}{1 + 2e^{-A} \cosh y}. \quad (\text{B.4})$$

In particular, y is the exact metric coordinate of f_A , and 0 is its analytic center. Since $V_A(y) \rightarrow \pm 1$ as $y \rightarrow \pm\infty$, the identity $\frac{d}{dy}f_A(X_A(y)) = V_A(y)$ proves blow-up at both endpoints. Also $Z_A(y) \rightarrow \pm\infty$, so f'_A takes every real value.

The exact parameter of f_A is $\nu_A = \sup_y V_A(y)^2$. For $0 \leq y \leq A$, $V_A(y) \leq y + 1 \leq A + 1$. For $y = A + u \geq A$, use $2e^{-A} \sinh y \leq 2e^{-A} \cosh y$ and $2e^{-A} \cosh y \geq e^u$ to get

$$V_A(y) \leq 1 + \frac{A + u - 1}{1 + e^u} \leq A + 1;$$

the final inequality follows from $u - 1 \leq Ae^u$ for $A > 4$. Oddness handles negative y . At $y = A/2$, the denominator in (B.4) is $1 + e^{-A/2} + e^{-3A/2} < 3/2$, so $V_A(A/2) > A/3$. Thus $A^2/9 < \nu_A \leq (A + 1)^2$. The speed V_A is not monotone: it exceeds one at $A/2$ but tends to one at the endpoint.

Fix r and $0 < \delta < 1/2$. Uniformly for $y \in [\delta A, (1 - \delta)A]$ as $A \rightarrow \infty$, $P_A(y) = 1 + o(1)$ and $Z_A(y) = y + o(1)$. Consequently, while the metric coordinate crosses this interval, the logarithmic central-path parameter increases by

$$\log Z_A((1 - \delta)A) - \log Z_A(\delta A) = \log \frac{1 - \delta}{\delta} + o(1).$$

Choose $B > \log((1 - \delta)/\delta)$ and weights with $\log w_i = B(r - i)$, and stop the central path at $s_1 = \log Z_A((1 - \delta)A)$. For large A , the r parameter windows, shifted by the weights, are disjoint, and in each one metric coordinate advances by $(1 - 2\delta)A$. Therefore the central arc has length at least $r(1 - 2\delta)A$.

At the final parameter, $Z_A(y_i) = Z_A((1 - \delta)A)e^{B(r - i)}$. For $y \geq \delta A$ and large A , $Z_A(y) \geq e^{y - A}/3$ and $Z_A((1 - \delta)A) < A$. Thus $y_i \leq A + \log(3A) + B(r - i)$, so the endpoint distance is at most $\sqrt{r}[A + \log(3A) + Br]$. Let $A \rightarrow \infty$ with r, δ, B fixed, then let $\delta \downarrow 0$. The ratio approaches $\sqrt{r}$, proving [proposition 7.2](#). This does not contradict comparisons that depend on the barrier parameter, such as (1.2): the scalar parameter, and hence the product parameter $r\nu_A$, diverges in this construction.

C Proofs for the concrete formulations

C.1 Grouped paraboloids and arbitrary PSD column packing

For a grouped tangent $(\dot{s}, \dot{w})$, put $\dot{q} = \dot{s} - 2\langle w, \dot{w} \rangle$. Direct differentiation gives

$$D^2(-\log q)[(\dot{s}, \dot{w}), (\dot{s}, \dot{w})] = (\dot{q}/q)^2 + 2\|\dot{w}\|^2/q. \quad (\text{C.1})$$

For a packed block, an arbitrary affine tangent gives $\dot{D} = \dot{S} - \dot{W}^T W - W^T \dot{W}$, and

$$D^2(-\log \det D)[(\dot{S}, \dot{W}), (\dot{S}, \dot{W})] = \text{tr}((D^{-1}\dot{D})^2) + 2\text{tr}(D^{-1}\dot{W}^T \dot{W}). \quad (\text{C.2})$$

These are Hessians in affine (s, w) or (S, W) coordinates; treating D as an independent affine variable when taking the second derivative would omit the second term. With c_ℓ columns in block ℓ , trace Cauchy–Schwarz gives

$$\|\dot{z}\|_F^2 \geq \sum_\ell \frac{1}{c_\ell} \left(\frac{d}{dt} \log \det D_\ell \right)^2. \quad (\text{C.3})$$

In the one-column case, the map to all $\log q$ does not increase length. Both Hessians are positive definite on the free affine coordinates. Standard self-concordance follows by restricting the classical

PSD barrier to the affine matrices $\begin{pmatrix} I & W \\ W^T & S \end{pmatrix}$, whose determinant is $\det D$; the one-column case gives (C.1). The derivative $DF = -\sum_\ell \text{tr}(D_\ell^{-1} \dot{D}_\ell)$ and (C.3) imply $|DF[\dot{z}]| \leq \sqrt{H} \|\dot{z}\|_F$, where $H = \sum_\ell c_\ell = kh$. This proves the upper bound H on the parameter; the fixed identity blocks contribute nothing.

For a feasible point, write $\delta_a = 1 - \langle c_a, w_a \rangle$. The diagonal residuals $d_\gamma = (D_{\ell(\gamma)})_{\gamma\gamma}$ satisfy

$$\sum_{\gamma \text{ from } a} d_\gamma = 1 - \|w_a\|^2 \leq 1 - \langle c_a, w_a \rangle^2 \leq 2\delta_a.$$

The same identity holds for the grouped residuals q_{aG} . If the objective gap is at most ε , then $\sum_a \delta_a \leq \varepsilon$ and

$$\prod_\ell \det D_\ell \leq \prod_\gamma d_\gamma \leq \prod_a (2\delta_a/h)^h \leq (2\varepsilon/H)^H. \quad (\text{C.4})$$

The first inequality is Hadamard's; it holds for all values of the completion variables, so the residuals need not be diagonal. Integrating (C.3) and using weighted Cauchy–Schwarz shows that every curve joining two points has length at least $H^{-1/2}$ times the absolute change in the sum of log determinants. If the reference point has $D_\ell = q_0 I$, (C.4) gives (10.8).

We now verify (10.7). At fixed W and fixed residual diagonals, Hadamard's inequality minimizes the barrier precisely when every residual D_ℓ is diagonal. Partial minimization over the unrestricted off-diagonal entries of S therefore leaves the grouped problem. Its equations in s_{aG} give one common residual q per ball, and its equations in w give $w_{aG} = (\tau q/2)c_{aG}$. Each ball's constraint $\sum_G s_{aG} = 1$ becomes $hq + (\tau q/2)^2 = 1$, proving (10.7). Positive definite Hessians make these stationary points the unique centers. Differentiated stationarity on the affine slice gives $H_\tau z'(\tau) = \nabla \ell$ and hence

$$\left\| \frac{dz}{d \log \tau} \right\|^2 = \tau^2 \frac{d}{d\tau} \ell(z(\tau)) = k\tau^2 r'(\tau) = H \left(1 - \frac{h}{\sqrt{h^2 + \tau^2}} \right).$$

At a center, $DF = \tau d\ell$ on the slice, so the same expression is the squared dual gradient norm. Its limit H is the exact parameter. Changing variables from τ to r gives the length element $\sqrt{H} \rho'(r) dr$, proving (10.12).

C.2 Norm trees: centers and distance

The positive-sheet domain (10.9) is a convex affine restriction of a product of Lorentz cones. Its Hessian is positive definite: any nonzero free direction changes at least one cone block, whose Hessian is positive definite. Restriction and addition preserve standard self-concordance. Summing a tree's residuals, every nonroot internal square cancels, and for each ball a ,

$$\sum_v q_{av} = 1 - \|w_a\|^2. \quad (\text{C.5})$$

The derivative in the coordinate t_{av} of a nonroot internal node v is $-2t_{av}/q_{av} + 2t_{av}/q_{a,\text{parent}(v)}$. Since $t_{av} > 0$, central stationarity forces all residuals in a tree to agree. Leaf stationarity gives $w_a = rc_a$, and (C.5) becomes $bq + r^2 = 1$, with $r = \tau q/2$. The positive internal coordinates are uniquely $t_{av} = (n_v q + r^2 C_{av})^{1/2}$; their root value is one. These equations prove feasibility and stationarity, and strict convexity proves uniqueness. They give the analytic center $w = 0$, $t_{av}^2 = n_v/b$ and the path and speed formulas of section 10.3.

For each Lorentz block, the parameter-two gradient inequality gives $(d \log q_{av})^2 \leq 2D^2(-\log q_{av})$ after restriction. Therefore the metric of F is at least $1/\sqrt{2}$ times the Euclidean metric in the $\log q_{av}$.

More sharply, with the exact restricted parameter ν proved below, the barrier itself is $\sqrt{\nu}$ -Lipschitz in metric length. Telescoping and the same AM–GM calculation over balls as in (C.4) give

$$\prod_{a,v} q_{av}(y) \leq (2\varepsilon/L)^L, \quad F(y) - F(\bar{z}) \geq L \log(k/(2\varepsilon)).$$

This proves (10.11). Differentiated stationarity gives the same speed as for grouped balls with $h = b, H = L$, so the same scalar integration proves (10.12). Thus the matching upper bound needs no asymptotic regularity of the KKT equations at the boundary.

C.3 Exact parameter after fixing the root

We prove (10.10) for one tree. Before fixing the root coordinate t , let $\Phi = -\sum_v \log q_v$ on the tree's cone and $H = \Phi''$. The barrier Φ is logarithmically homogeneous of degree $2b$; hence $H z = -\Phi'(z)$ and $\|\Phi'\|_{H^{-1}}^2 = 2b$. Let e_t be the root-coordinate covector. Orthogonal projection of the gradient onto $\{h : e_t^T h = 0\}$ gives the exact restricted norm

$$\|DF\|_{F,*}^2 = 2b - \frac{t^2}{e_t^T H^{-1} e_t}, \quad \nu_T = 2b - \frac{1}{\sup_z \beta_T(z)}, \quad \beta_T(z) = \frac{e_t^T H^{-1} e_t}{t^2}. \quad (\text{C.6})$$

Since the Dikin ellipsoid lies in the halfspace $t > 0$, $e_t^T H^{-1} e_t \leq t^2$, so $\beta_T \leq 1$. Indeed, if the inequality failed, a vector of local norm less than one could move the root coordinate to zero.

If a child of the root is a leaf, set $t = 1$ and that leaf to $\sqrt{1 - \delta}$. Set other root leaves to zero and scale every internal child subtree from a fixed interior point by δ^2 . The root residual is $q = \delta - O(\delta^4) > 0$. For the full-space trial direction $h_t = 1$, $h_{\text{leaf}} = 1$, and every other component zero, descendant blocks contribute zero, while

$$D^2\Phi[h, h] = \frac{\{2(1 - \sqrt{1 - \delta})\}^2}{q^2} \rightarrow 1.$$

Here the root residual's second derivative in that direction is zero. The variational identity $(e_t^T H^{-1} e_t)^{-1} = \min_{h_t=1} h^T H h$ now gives $\sup \beta_T = 1$. This includes a one-node tree without descendants. This proves $\nu_T = 2b - 1$ with an explicit feasible sequence.

Suppose instead that all root children are internal, with positive axial coordinates $a_1, \dots, a_c$. Eliminate all their strict descendants from the Hessian. Each descendant appears only in the barrier terms of its own child subtree, so the reduced Hessian in (t, a) is the Hessian of the root term plus $\text{diag}(0, h_1, \dots, h_c)$. Here the effective axial curvature h_i is the Schur complement on a_i of the Hessian of child subtree i 's barrier terms. By the same Dikin halfspace argument on that subtree, $h_i \geq a_i^{-2}$. Increasing an axial curvature increases the root Schur complement, so a lower bound is obtained from

$$-\log(t^2 - \sum_i a_i^2) - \sum_i \log a_i.$$

Normalize $t = 1$, put $r = \sum_i a_i^2$, $q = 1 - r$, and $Z = \sum_i a_i^4/(q + 2a_i^2)$. Its Hessian in the child coordinates is a diagonal matrix with entries $2/q + a_i^{-2}$ plus $4aa^T/q^2$; the root–child row is $-4a^T/q^2$, and the root entry is $2(1 + r)/q^2$. The rank-one inverse formula therefore gives the root Schur complement

$$S = \frac{2(1 + r) - 16Z/(q + 4Z)}{q^2}. \quad (\text{C.7})$$

Since $x/(q + 2x)$ increases in $x \geq 0$, $\sum_i a_i^4/(q + 2a_i^2) \leq r^2/(q + 2r) = r^2/(1 + r)$. The inequality $S \geq 2$ is equivalent to $r(3 - r) \geq 4(2 - r)Z$; replacing Z by this upper bound leaves $3r(1 - r)^2/(1 + r) \geq 0$.

Thus $\beta_T \leq 1/2$. Conversely, scale all child subtrees by $\delta \downarrow 0$. The trial direction $h_t = 1$, with all other variables fixed, has norm squared $2(1+r)/(1-r)^2 \rightarrow 2$. Together with the lower bound on the minimizing Schur complement this proves $\sup \beta_T = 1/2$ and $\nu_T = 2b - 2$. For independent trees, the squared gradient norms and their suprema add, proving the product parameter used in (10.11).

C.4 Unequal group counts and objective weights

In all three formulations, let ball a have m_a groups or internal nodes, let $M_0 = \sum_a m_a$, $p_a = m_a/M_0$, and maximize $\sum_a \lambda_a \langle c_a, w_a \rangle$ with every $\lambda_a > 0$. Define the scale

$$\Delta_{\text{eff}} = \prod_a (\lambda_a/p_a)^{p_a}. \quad (\text{C.8})$$

At the analytic center, computed ball by ball, every residual of ball a equals $1/m_a$. For an accurate point y , put $e_a = \lambda_a(1 - \langle c_a, w_a \rangle)$, so $\sum_a e_a \leq \varepsilon$. Telescoping, or the residual diagonal identities and Hadamard's inequality, show that the product of the residuals at y divided by that at the analytic center is at most $\prod_a (2e_a/\lambda_a)^{m_a}$. This product is maximized at $e_a = \varepsilon p_a$, giving

$$F(y) - F(\bar{z}) \geq M_0 \log(\Delta_{\text{eff}}/(2\varepsilon)).$$

Consequently the distance from the analytic center to the accurate set is at least

$$\frac{M_0}{\sqrt{\nu}} [\log(\Delta_{\text{eff}}/(2\varepsilon))]_+, \quad \nu = \begin{cases} M_0, & \text{grouped or packed,} \\ \sum_a \nu_{T_a}, & \text{norm trees.} \end{cases} \quad (\text{C.9})$$

For grouped balls and packings, the parameter M_0 remains exact by the central-path calculation for each ball, with parameter $\tau\lambda_a$. For equal weights $\lambda_a = \lambda > 0$, $\Delta_{\text{eff}} = \lambda \exp(-\sum_a p_a \log p_a)$; weighted AM–GM gives $\Delta_{\text{eff}} \leq \sum_a \lambda_a$ with equality precisely when λ_a is proportional to m_a . Thus the scale Δ_{eff} , not only the count M_0 , matters for movement.

References

- [1] Xavier Allamigeon, Pascal Benchimol, Stéphane Gaubert, and Michael Joswig. Log-barrier interior point methods are not strongly polynomial. *SIAM Journal on Applied Algebra and Geometry*, 2(1):140–178, 2018. doi: 10.1137/17M1142132.
- [2] Xavier Allamigeon, Stéphane Gaubert, and Nicolas Vandame. No self-concordant barrier interior point method is strongly polynomial. In *Proceedings of the 54th Annual ACM SIGACT Symposium on Theory of Computing*, pages 515–528. Association for Computing Machinery, 2022. doi: 10.1145/3519935.3519997.
- [3] Xavier Allamigeon, Daniel Dadush, Georg Loho, Bento Natura, and László A. Végh. Interior point methods are not worse than Simplex. *SIAM Journal on Computing*, 54(5):FOCS22–178–FOCS22–264, 2025. doi: 10.1137/23M1554588.
- [4] Michel Baes. Convexity and differentiability properties of spectral functions and spectral mappings on Euclidean Jordan algebras. *Linear Algebra and its Applications*, 422(2–3):664–700, 2007. doi: 10.1016/j.laa.2006.11.025.

- [5] Sébastien Bubeck and Ronen Eldan. The entropic barrier: Exponential families, log-concave geometry, and self-concordance. *Mathematics of Operations Research*, 44(1):264–276, 2019. doi: 10.1287/moor.2017.0923.
- [6] Jordi Castro and Jordi Cuesta. Quadratic regularizations in an interior-point method for primal block-angular problems. *Mathematical Programming*, 130(2):415–445, 2011. doi: 10.1007/s10107-010-0341-2.
- [7] Sinho Chewi. The entropic barrier is n -self-concordant. In Ronen Eldan, Bo’az Klartag, Alexander Litvak, and Emanuel Milman, editors, *Geometric Aspects of Functional Analysis*, volume 2327 of *Lecture Notes in Mathematics*, pages 209–222. Springer, Cham, 2023. doi: 10.1007/978-3-031-26300-2_6.
- [8] Daniel Dadush, Haoyuan Ma, Bento Natura, and László A. Végh. Trust region interior point methods: Optimal ℓ_2 - and faster wide-neighborhood path following. In *Proceedings of the 58th Annual ACM Symposium on Theory of Computing*, pages 755–766. Association for Computing Machinery, June 2026. doi: 10.1145/3798129.3800790.
- [9] Antoine Deza, Eissa Nematollahi, and Tamás Terlaky. How good are interior point methods? Klee–Minty cubes tighten iteration-complexity bounds. *Mathematical Programming*, 113:1–14, 2008. doi: 10.1007/s10107-006-0044-x.
- [10] Jacques Faraut and Adam Korányi. *Analysis on Symmetric Cones*. Oxford Mathematical Monographs. Clarendon Press, Oxford, 1994. doi: 10.1093/oso/9780198534778.001.0001.
- [11] João Gouveia, Pablo A. Parrilo, and Rekha R. Thomas. Lifts of convex sets and cone factorizations. *Mathematics of Operations Research*, 38(2):248–264, 2013. doi: 10.1287/moor.1120.0575.
- [12] Raphael A. Hauser and Osman Güler. Self-scaled barrier functions on symmetric cones and their classification. *Foundations of Computational Mathematics*, 2(2):121–143, 2002. doi: 10.1007/s102080010022.
- [13] Roland Hildebrand. A lower bound on the barrier parameter of barriers for convex cones. *Mathematical Programming*, 142:311–329, 2013. doi: 10.1007/s10107-012-0576-1.
- [14] Hiroshi Hirai, Harold Nieuwboer, and Michael Walter. Interior-point methods on manifolds: Theory and applications. *Foundations of Computational Mathematics*, 2026. doi: 10.1007/s10208-026-09756-8. Published online.
- [15] Yin Tat Lee and Man-Chung Yue. Universal barrier is n -self-concordant. *Mathematics of Operations Research*, 46(3):1129–1148, 2021. doi: 10.1287/moor.2020.1113.
- [16] Lucas Lévy, Jean-Lou Valeau, Arya Akhavan, and Patrick Rebeschini. Self-concordant perturbations for linear bandits, 2026. [arXiv:2510.24187v3](https://arxiv.org/abs/2510.24187v3).
- [17] Adrian S. Lewis and Hristo S. Sendov. Twice differentiable spectral functions. *SIAM Journal on Matrix Analysis and Applications*, 23(2):368–386, 2001. doi: 10.1137/S089547980036838X.
- [18] George G. Lorentz. On the theory of spaces Λ . *Pacific Journal of Mathematics*, 1(3):411–429, 1951. doi: 10.2140/pjm.1951.1.411.
- [19] Yurii Nesterov and Arkadi Nemirovski. Primal central paths and Riemannian distances for convex sets. *Foundations of Computational Mathematics*, 8(5):533–560, 2008. doi: 10.1007/s10208-007-9019-4.

- [20] Yurii Nesterov and Arkadii Nemirovskii. *Interior-Point Polynomial Algorithms in Convex Programming*, volume 13 of *SIAM Studies in Applied Mathematics*. Society for Industrial and Applied Mathematics, Philadelphia, 1994. doi: 10.1137/1.9781611970791.
- [21] Yurii E. Nesterov and Michael J. Todd. On the Riemannian geometry defined by self-concordant barriers and interior-point methods. *Foundations of Computational Mathematics*, 2(4):333–361, 2002. doi: 10.1007/s102080010032.
- [22] Atsumi Ohara, Hideyuki Ishi, and Takashi Tsuchiya. Doubly autoparallel structure and curvature integrals: Applications to iteration complexity for solving convex programs. *Information Geometry*, 7:555–586, 2024. doi: 10.1007/s41884-023-00116-x.
- [23] Erik Alex Papa Quiroz and Paulo Roberto Oliveira. New self-concordant barrier for the hypercube. *Journal of Optimization Theory and Applications*, 135(3):475–490, 2007. doi: 10.1007/s10957-007-9220-2.
- [24] Frank Permenter. A geodesic interior-point method for linear optimization over symmetric cones. *SIAM Journal on Optimization*, 33(2):1006–1034, 2023. doi: 10.1137/20M1385019.